Online Appendix: Public Service Decline and Support for the Populist Right

Evidence from England's National Health Service

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Contents

GP Practice Closures 4

Data Merging and Treatment Assignment 7
- GP Patient Survey . . . 7
- Understanding Society . . . 8
- British Election Study Panel . . . 8
- Estimation . . . 9

GP Patient Survey Analysis 10
- Fixed Effects Estimates . . . 10
- Matrix Completion . . . 12
- Parallel Trends Sensitivity Analysis . . . 14
- Alternative Treatment Assignment: Nearest GP Practice Closure . . . 17

Understanding Society Analysis 21
- Matrix Completion Estimates . . . 25

British Election Study Analysis 26
- Fixed Effects Estimates . . . 26
- Matrix Completion Estimates . . . 29

Topic Model 30

Moderation Analysis 31
- Populist Radical Right . . . 31
- Labour Party . . . 31
- Green Party . . . 32
- Liberal Democrats . . . 33
- Conservative Party . . . 33

(Re-)Opening Practices Robustness Check 35

Linking the NHS with Immigration in Party Communication 36
- Dictionaries and connectors . . . 36
- YouTube transcripts: windowing and measures . . . 36
- UKIP press releases: sentence-based design . . . 37
- Interpretation . . . 38
- Extended Results . . . 40

Placebo Test: Reported *Past* Vote 42

Robustness Check: Not-yet-treated as the Control Group 43

Placebo Tests 45

Treatment Schedule 46

Effects of Closures on other Parties 47

Effects of Closures on Moving **50**

Vote Intention vs. Actual Vote Share **51**

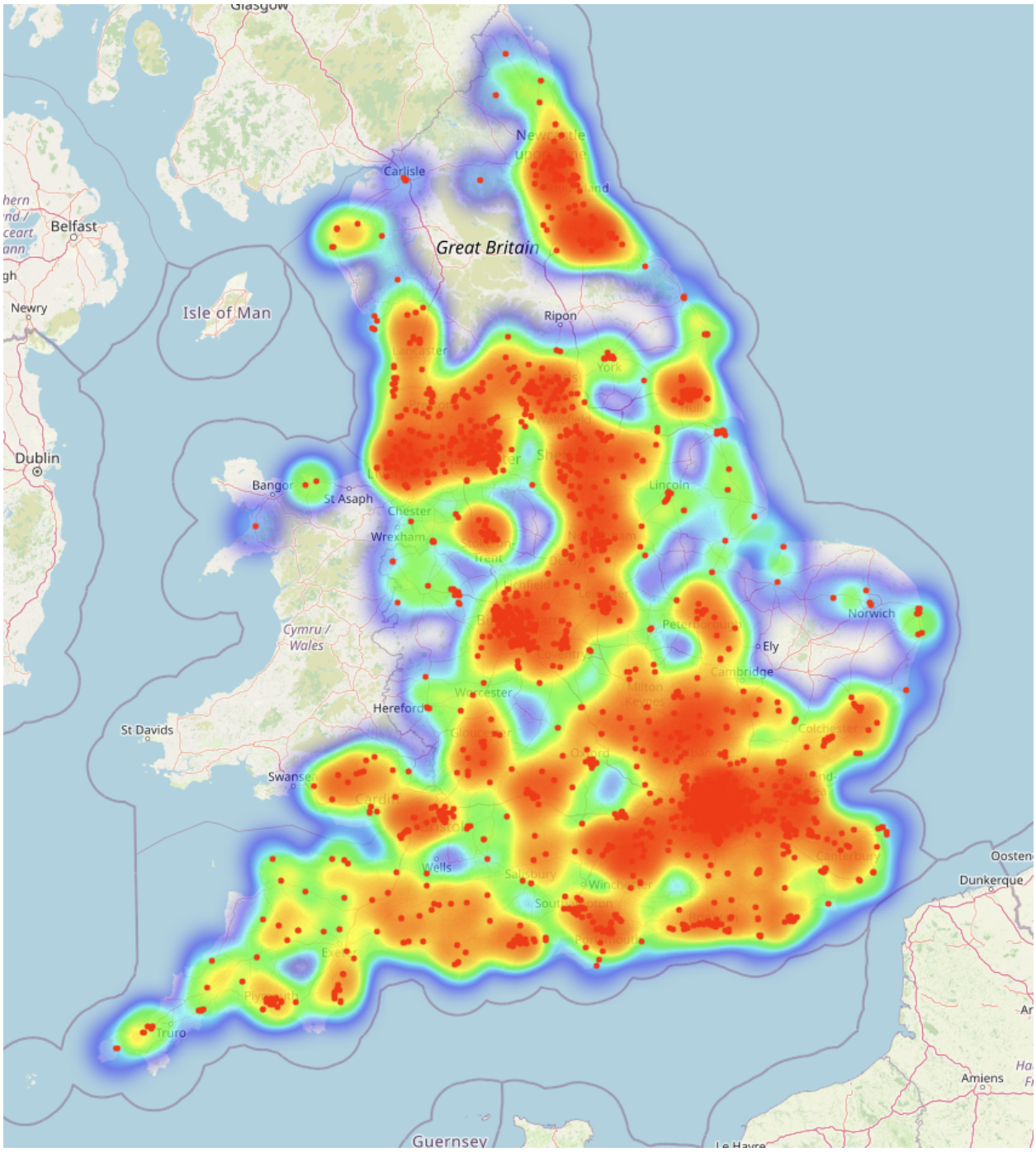
GP Practice Closures

GP practices can close for a number of reasons. While multiple practices that close are simply re-opened under new management and a new staffing team, many others close permanently. In either case, a disruption of care can have a negative effect on patients. Although we are unaware of a data source that provides details on why a given practice closes, a detailed investigation by Pulse, a GP publication (Carter and Kaffash 2022), of 474 different practices that had permanently closed without replacement found that the most common reason for closure was staffing issues. The Pulse team found that the most common reason were as follows:

1. Staffing Shortages (42%)
2. GP Retirement (20%)
3. Inability to Recruit (10%)
4. GP Resignation (8%)
5. Non-renewal of an APMS (Alternative Provider Medical Services) contract (8%)
6. Failure to Meet CQC (Care Quality Commission) Standards (6%)

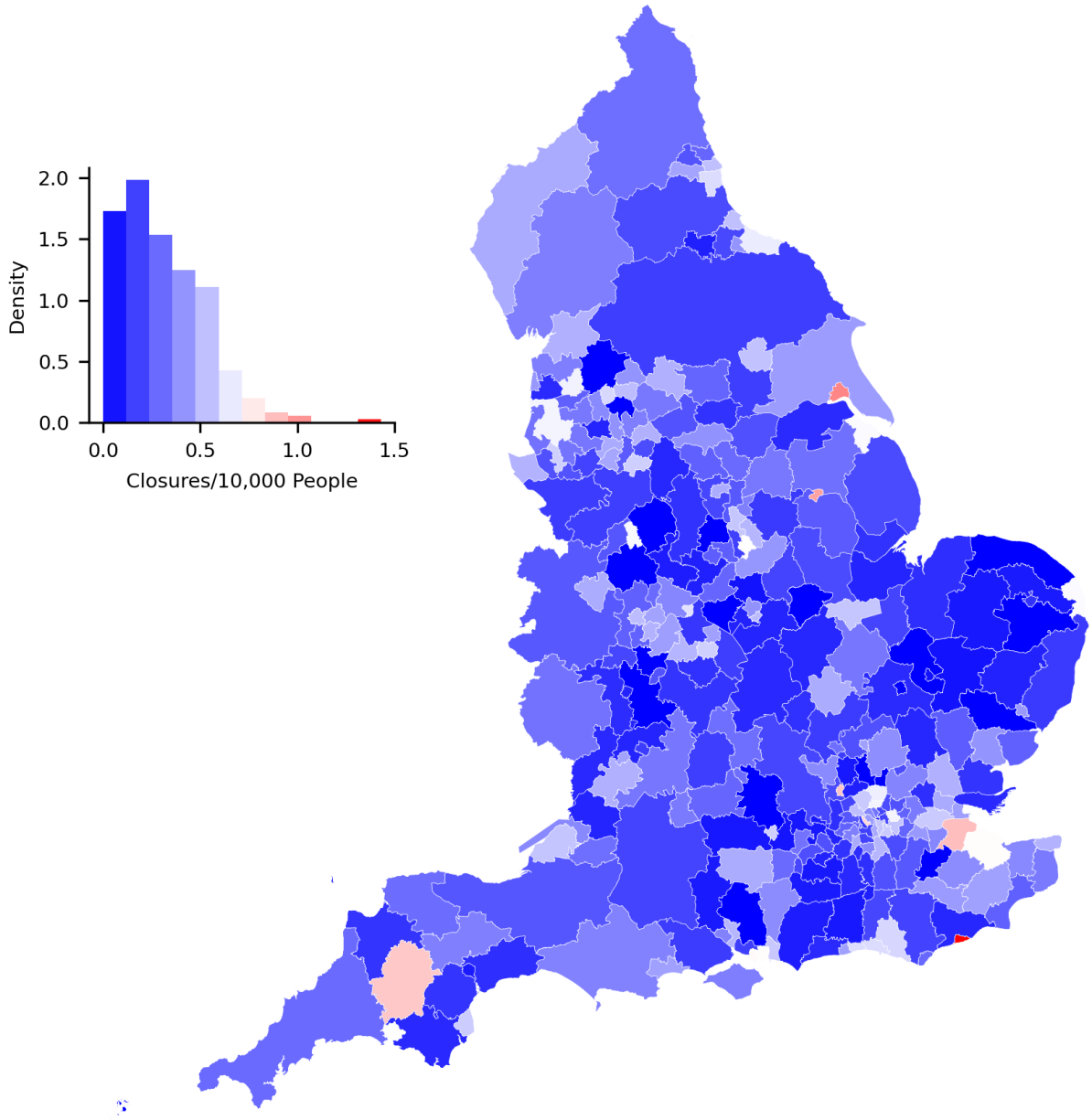
In our analysis, we assign exposure to a closure from the date at which a given practice no longer has a license to practice according to NHS England. This date is available from NHS England and (NHS England 2023).

Figure A1: GP Practice Closures



Note: Locations of GP practice closures in England between 2013 and 2022.

Figure A2: GP Practice Closures/Local Authority Population



Note: Number of GP practice closures per 10k population in each local authority. The figure uses 2021 population estimates for local authorities. All GP practice closures between 2013 and 2023 in England are included.

Data Merging and Treatment Assignment

In addition to the USOC and BES data, both of which we merge with the GP practice closure data, we augment our datasets with contextual variables that allow us to examine the effect heterogeneity with respect to a range of relevant moderators. To test the moderating effect of immigration, we consider international migration into a given local authority area. We rely on data made available by the Office for National Statistics (2023) on the rate of immigration in local authorities, which we measure on a per-resident basis. Moreover, we collect data on the number of migrants registering at a GP practice for the first time. We use the annual rate of new migrant registrations (per resident population) for each local authority (Office for National Statistics 2023).

Below, we discuss the treatment assignment procedures for the GP closures dataset with the GP Patient Survey, USOC and the BES panel. In all cases, treatment assignment occurs using the most granular geographic indicator available in the dataset. We additionally explore alternative treatment assignment procedures to ensure robustness.

GP Patient Survey

Treatment Assignment 1: Middle Layer Super Output Area (MSOA) and date.

The GP Patient Survey contains the practice codes of each GP practice for which a patient is registered. We use the practice codes to locate the MSOA of each practice. We then merge the GP patient survey data with the GP closures data on the MSOA of the practice. We consider all practices in a given MSOA as treated in the event that a practice in that MSOA has closed from the point of closure onwards. The 'treatment group' would therefore include patients who are registered to any practice that shares an MSOA with a practice that has closed. This ensures that the estimates include patients who are registered at practices that absorbs the patients of a local practice that has closed.

Treatment Assignment 2: Nearest GP practice and date.

As a robustness check, we implement an alternative treatment assignment based on geographic proximity rather than MSOA membership. For each practice, we identify the nearest other GP practice that closed within the panel window (2012–2023) using the Euclidean distance between practice latitude–longitude coordinates. A practice is considered treated from the year in which that nearest practice closes. Practices for which no closure occurred within the panel window are assigned to the never-treated group. Relative to the MSOA-based assignment, this approach is more targeted: it identifies the single most geographically proximate closure for each practice, regardless of administrative boundaries, and therefore most directly captures the patient displacement mechanism whereby displaced patients register at the closest available alternative. The results of this robustness check are presented in section .

Understanding Society

Treatment Assignment 1: $< 2km$ from closure and date

Treatment Assignment 2: Postcode sector and date

The Understanding Society (USOC) dataset provides Ordnance Survey grid reference coordinates for each respondent, expressed as Easting (E_i) and Northing (N_i) values in metres within the British National Grid (BNG) system. To align the data with other sources, we round each coordinate to the nearest 1,000 metres (approximately 1 km). Formally, for each observation i:

$$E_i^{(r)} = 1000 \times \text{round}\left(\frac{E_i}{1000}\right), \quad N_i^{(r)} = 1000 \times \text{round}\left(\frac{N_i}{1000}\right),$$

where round$(\cdot)$ denotes rounding to the nearest integer. For example, an original Easting value of 451,528 is rounded to 452,000. This operation groups all locations within the same 1 km $\times$ 1 km BNG grid cell to a common centroid, such that any two points sharing identical $(E_i^{(r)}, N_i^{(r)})$ values are at most $\sqrt{2}$ km ($\sim$1.41km) apart.

We apply the same rounding procedure to the GP Closures dataset, first converting GP practice postcode centroids to BNG coordinates. The two datasets are then matched by the rounded Easting and Northing values, effectively linking respondents to GP practices located within the same 1 km grid cell.

As a robustness check, we implement an alternative matching approach based on postcode geography. Specifically, we convert each respondent's BNG coordinates to postcode units using lookup data from the UK Office for National Statistics (ONS), and then match records at the postcode *sector* level, defined by the first five characters of the postcode (e.g., "AB12 3"). Each postcode sector typically contains between 500 and 3,000 addresses. Treatment assignment based on postcode sector yields results that are nearly identical to those obtained using the 1 km grid-based matching approach, supporting the robustness of our spatial linkage method.

Two waves in the USOC dataset do not ask about which party respondents would support if the election was held the next day (waves D and H). We drop both waves in our analysis.

British Election Study Panel

Treatment Assignment: Middle Layer Super Output Area (MSOA) and date.

The most granular geographic identifier in the British Election Study is the UK Census Middle Layer Super Output Area (MSOA). There are 7,264 MSOAs in England and Wales, containing approximately 7,000–10,000 individuals in each. We merge the GP Practice data to ONS data that links each UK postcode to respective Census geographic identifiers, including MSOA, and then merge the BES data to the GP Closures data on MSOA.

Estimation

For reference throughout this appendix, we reproduce our primary event-study specification from the main text:

$$Y_{i,t} = \alpha_g + \lambda_t + \zeta_i + \sum_{\ell \neq -1} \mu_\ell 1\{t - E_i = \ell\} + X'_{g,t}\beta + \varepsilon_{i,g,t} \tag{A1}$$

where $Y_{i,t}$ is the outcome of interest for unit i (a GP practice or an individual respondent) at time t, E_i denotes the period of first exposure to a GP practice closure, and the coefficients μ_ℓ trace the effect of exposure at each relative time period ℓ, with $\ell = -1$ (the year prior to exposure) serving as the omitted baseline. The vector $X_{g,t}$ includes time-varying contextual covariates measured annually at the local authority level.

GP Patient Survey Analysis

The full results of the analysis of the GP Patient Survey data – including all robustness checks – are provided in this section.

Fixed Effects Estimates

The following tables present Sun and Abraham (2021) event-study estimates for both GPPS outcomes across four model specifications, progressively adding control variables. All models include LAD and year fixed effects with two-way clustered standard errors (LAD × year). Model 1 includes no controls; Model 2 adds the IMD Score; Model 3 adds the unemployment rate; Model 4 adds migration controls (international migration inflow proportion and migrant GP registrations proportion). Event-study dynamics are shown in the figures in section .

Table A1: Effects of GP Practice Closures on Positive Overall Experience (GPPS)

Dependent Variable:	Positive overall experience with GP practice			
Model:	(1)	(2)	(3)	(4)
Variables				
ATT	-0.0175***	-0.0174***	-0.0164***	-0.0149***
	(0.0026)	(0.0026)	(0.0027)	(0.0024)
IMD Score		-0.0003	-0.0008	-0.0010**
		(0.0005)	(0.0005)	(0.0004)
Unemployment Rate			0.0039***	0.0035***
			(0.0011)	(0.0008)
International Migration (inflow)				-0.0064
				(0.0057)
Migrant GP Registrations				0.0050
				(0.0038)
Fixed-effects				
LAD	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	87,111	85,838	77,938	66,282
R^2	0.37624	0.37875	0.37743	0.21114
Within R^2	0.00684	0.00681	0.00819	0.00763

Clustered (LAD & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Two-way clustered (LAD × year) standard errors in parentheses. All models include LAD and year fixed effects. ATT is the Sun and Abraham (2021) aggregated average treatment effect on the treated. Model 1: no controls. Model 2: adds IMD Score. Model 3: adds Unemployment Rate. Model 4: adds migration controls (international migration inflow proportion and migrant GP registrations proportion).

Table A2: Effects of GP Practice Closures on Negative Appointment Experience (GPPS)

Dependent Variable:	Negative overall experience making an appointment			
Model:	(1)	(2)	(3)	(4)
Variables				
ATT	0.0131***	0.0129***	0.0119***	0.0100***
	(0.0026)	(0.0026)	(0.0026)	(0.0023)
IMD Score		0.0002	0.0008	0.0010**
		(0.0007)	(0.0007)	(0.0004)
Unemployment Rate			-0.0045***	-0.0038***
			(0.0013)	(0.0008)
International Migration (inflow)				0.0073
				(0.0057)
Migrant GP Registrations				-0.0007
				(0.0045)
Fixed-effects				
LAD	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	87,037	85,764	77,880	66,278
R^2	0.35095	0.35294	0.35371	0.18689
Within R^2	0.00531	0.00530	0.00720	0.00665

Clustered (LAD & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Two-way clustered (LAD × year) standard errors in parentheses. All models include LAD and year fixed effects. ATT is the Sun and Abraham (2021) aggregated average treatment effect on the treated. Model 1: no controls. Model 2: adds IMD Score. Model 3: adds Unemployment Rate. Model 4: adds migration controls (international migration inflow proportion and migrant GP registrations proportion).

Matrix Completion

Table A3: Dynamic Effects of Closures on Positive Overall Experience at Practice

Time	ATT	S.E.	p.value
-5	-0.002119	0.002538	0.403738
-4	0.001568	0.002026	0.438867
-3	0.000496	0.001733	0.774585
-2	-0.000078	0.001605	0.961051
-1	-0.000139	0.001817	0.939094
0	-0.005508	0.002471	0.025784
1	-0.011274	0.003502	0.001285
2	-0.012274	0.004173	0.003267
3	-0.011556	0.005413	0.032794
4	-0.013487	0.005424	0.012894
5	-0.019215	0.006391	0.002643

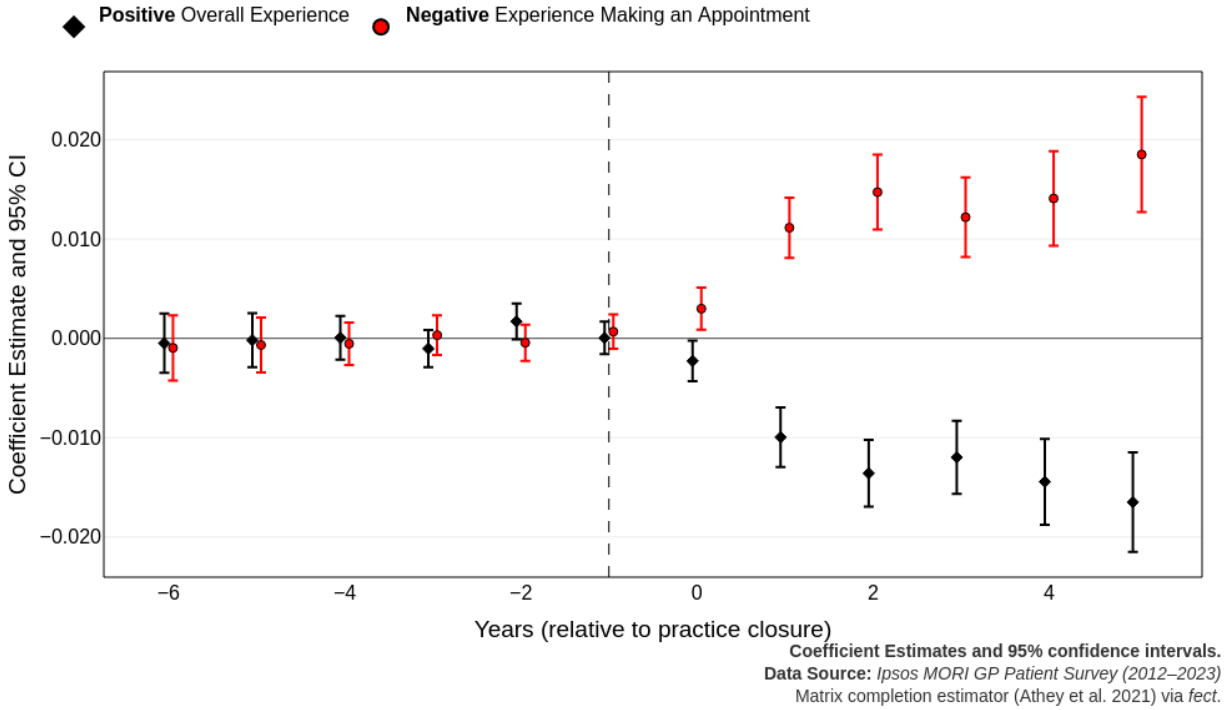
Note: Dynamic matrix completion ATT estimates for the effects of practice closures on patient attitudes. The outcome variable is the percentage of patients who report that their overall experience at the GP practice is *very good*. Estimates are made using the `Fect` library in R (Liu, Wang, and Xu 2022).

Table A4: Dynamic Effects of Closures on Negative Overall Experience Making an Appointment at Practice

Time	ATT	S.E.	p.value
-5	-0.000818	0.002424	0.735831
-4	-0.002106	0.002259	0.351114
-3	-0.001119	0.001936	0.563364
-2	0.000545	0.001940	0.778662
-1	-0.000478	0.001942	0.805567
0	0.005006	0.001955	0.010459
1	0.008265	0.003093	0.007542
2	0.010985	0.004064	0.006866
3	0.010528	0.004402	0.016758
4	0.015532	0.005025	0.001996
5	0.022412	0.005988	0.000182

Note: Dynamic matrix completion ATT estimates for the effects of practice closures on patient attitudes. The outcome variable is the percentage of patients who report that their experience making an appointment at the practice is *very poor*. Estimates are made using the `Fect` library in R (Liu, Wang, and Xu 2022).

Figure A3: Effect of GP Practice Closures on Patient Experiences in Area Practices (MC)



Note: Dynamic estimates for the effects of GP practice closures on patient overall satisfaction in GP practices after a closure. The outcome variables are the percentage of patients at a practice who report a positive overall experience at the GP practice (black circles) and the percentage of patients who report a negative overall experience making an appointment at the GP practice (red circles).

Parallel Trends Sensitivity Analysis

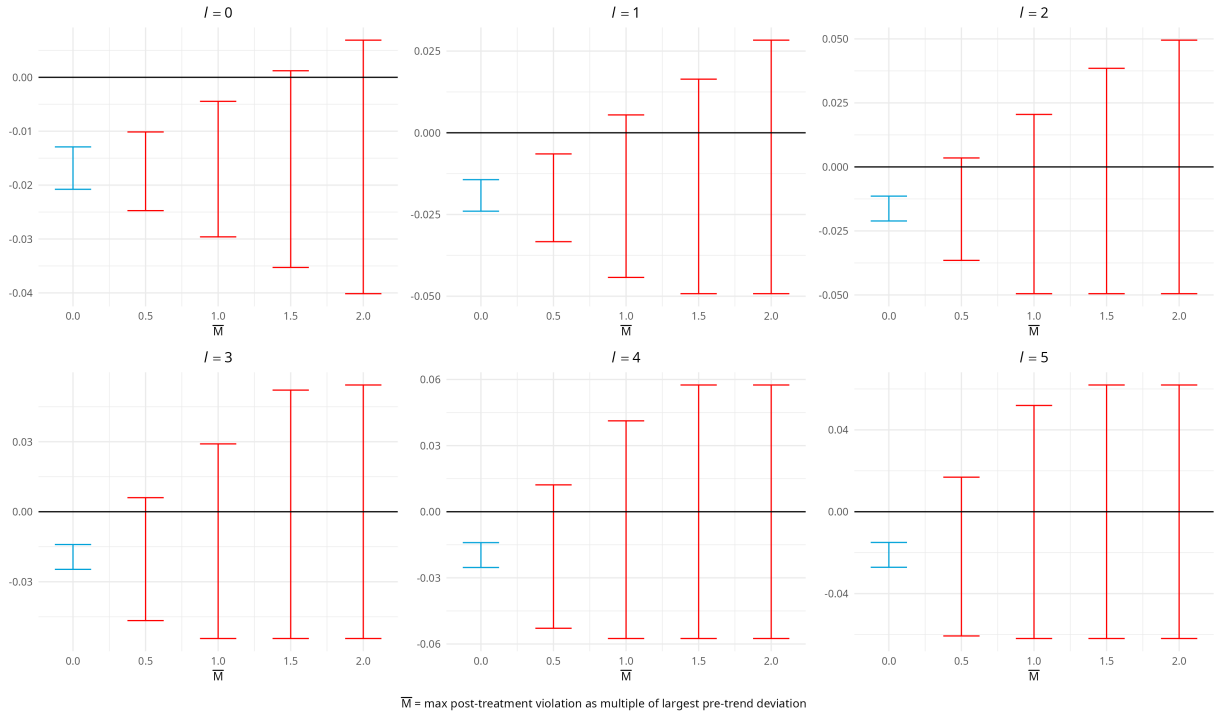
The fixed-effects event study estimates for the GP Patient Survey (GPPS) outcomes presented in section show some evidence of pre-treatment violations: for both outcomes, the event-study coefficients for periods $\ell = -4$, $\ell = -3$, and $\ell = -2$ are statistically significant, although small in magnitude relative to the post-treatment effects. To assess whether the main findings are sensitive to these violations, we apply the sensitivity analysis framework of Rambachan and Roth (2023), implemented via the `HonestDiD` package in `R`.

The framework asks how large a violation of parallel trends would need to be—relative to the observed pre-trend deviations—to overturn the post-treatment conclusions. We use the *relative magnitude* restriction (Δ^{RM}), which bounds the maximum post-treatment deviation from parallel trends as a multiple $\bar{M}$ of the largest observed pre-treatment deviation. $\bar{M} = 1$ is the natural benchmark: violations cannot exceed the size of the observed pre-trends. We construct uniformly valid 95% confidence sets using the *conditional least-favorable* (C-LF) method.

The analysis is conducted separately for the two GPPS outcomes (positive overall experience at the GP practice and negative experience making an appointment). We apply the analysis to each post-treatment period from $\ell = 0$ to $\ell = 5$ individually. All available pre-treatment periods are used to calibrate the sensitivity bounds.

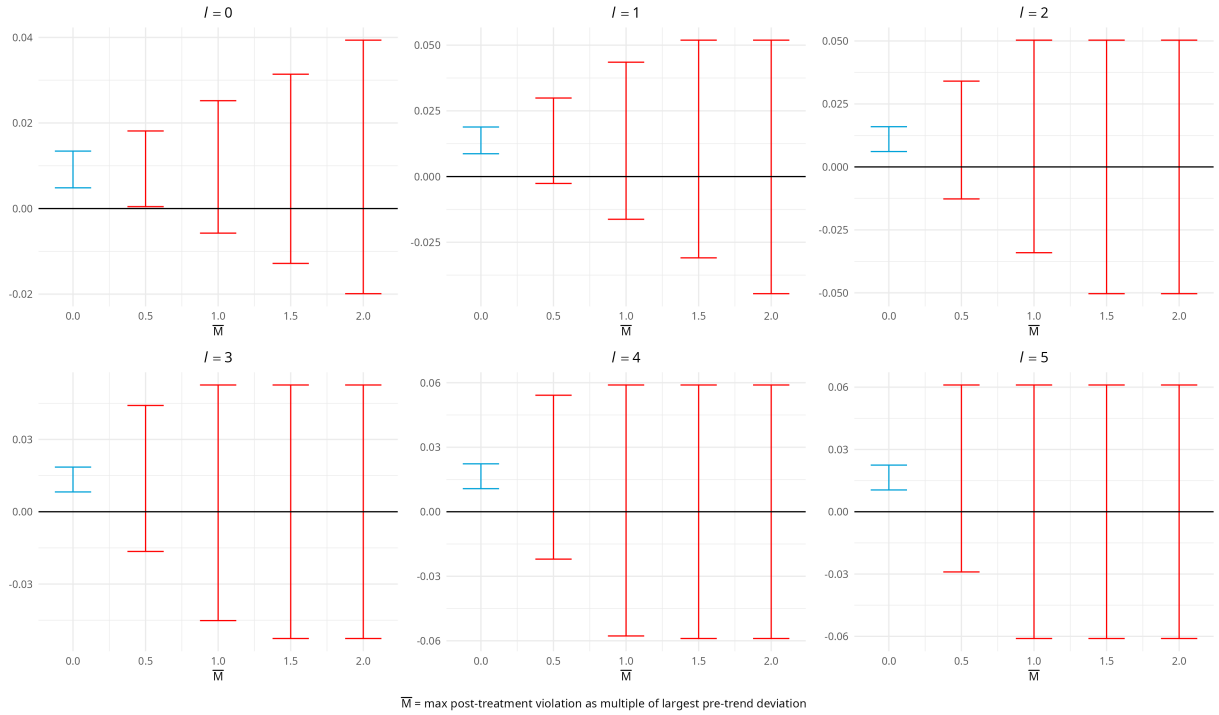
Figure A4 and Figure A5 present the results for all six post-treatment periods. The original 95% confidence intervals at $\ell = 0$ are $[-0.021, -0.013]$ for positive experience and $[+0.005, +0.013]$ for negative appointment experience—both entirely away from zero. For positive overall experience, the robust CI excludes zero at $\bar{M} = 0.50$ for the first two periods. For negative appointment experience, the robust CI excludes zero at $\bar{M} = 0.50$ for $\ell \geq 1$ and at $\bar{M} = 1.00$ at $\ell = 0$. For positive overall experience, the robust CI excludes zero at $\bar{M} = 0.50$ for the first period, while additional periods are more sensitive to potential trend violations.

Figure A4: Relative Magnitude Sensitivity: Positive Overall Experience, $\ell = 0$ to $\ell = 5$



Note: Each panel shows the relative-magnitude (Δ^{RM}) sensitivity analysis for a single post-treatment period, $\ell = 0$ through $\ell = 5$. The shaded region is the robust 95% confidence set; the point and whiskers at $\bar{M} = 0$ show the original (uncorrected) confidence interval. Estimated using the C-LF method via the `HonestDiD` package (Rambachan and Roth 2023).

Figure A5: Relative Magnitude Sensitivity: Negative Appointment Experience, $\ell = 0$ to $\ell = 5$



Note: Each panel shows the relative-magnitude (Δ^{RM}) sensitivity analysis for a single post-treatment period, $\ell = 0$ through $\ell = 5$. The shaded region is the robust 95% confidence set; the point and whiskers at $\bar{M} = 0$ show the original (uncorrected) confidence interval. Estimated using the C-LF method via the `HonestDiD` package (Rambachan and Roth 2023).

Alternative Treatment Assignment: Nearest GP Practice Closure

In the primary GPPS analysis, treatment is assigned at the Middle Layer Super Output Area (MSOA) level: a practice is considered treated from the year in which *any* practice in its MSOA closes. This captures both the direct displacement of patients to surviving practices within the same small area and the broader pressure on local GP capacity that follows a closure.

As a robustness check, we adopt a more targeted treatment assignment based on geographic proximity. For each practice i, we identify its nearest neighbour among all practices that closed within the panel window (2012–2023):

$$j^*(i) \;=\; \arg\min_{j \neq i,\, j \in \mathcal{C}} \; d(i, j),$$

where $\mathcal{C}$ is the set of practices that closed during the panel and $d(\cdot, \cdot)$ is the Euclidean distance computed from practice latitude–longitude coordinates. The treatment cohort variable G_i^{nearest} is then set equal to the closure year of $j^*(i)$. Practices for which no closure occurred within the panel window are assigned $G_i^{\text{nearest}} = 10{,}000$ (the never-treated placeholder used throughout). The rationale for this assignment is that the nearest-practice closure most directly determines which surviving practice absorbs displaced patients: when $j^*(i)$ closes, practice i is the most geographically proximate alternative and therefore experiences the largest influx of newly registered patients.

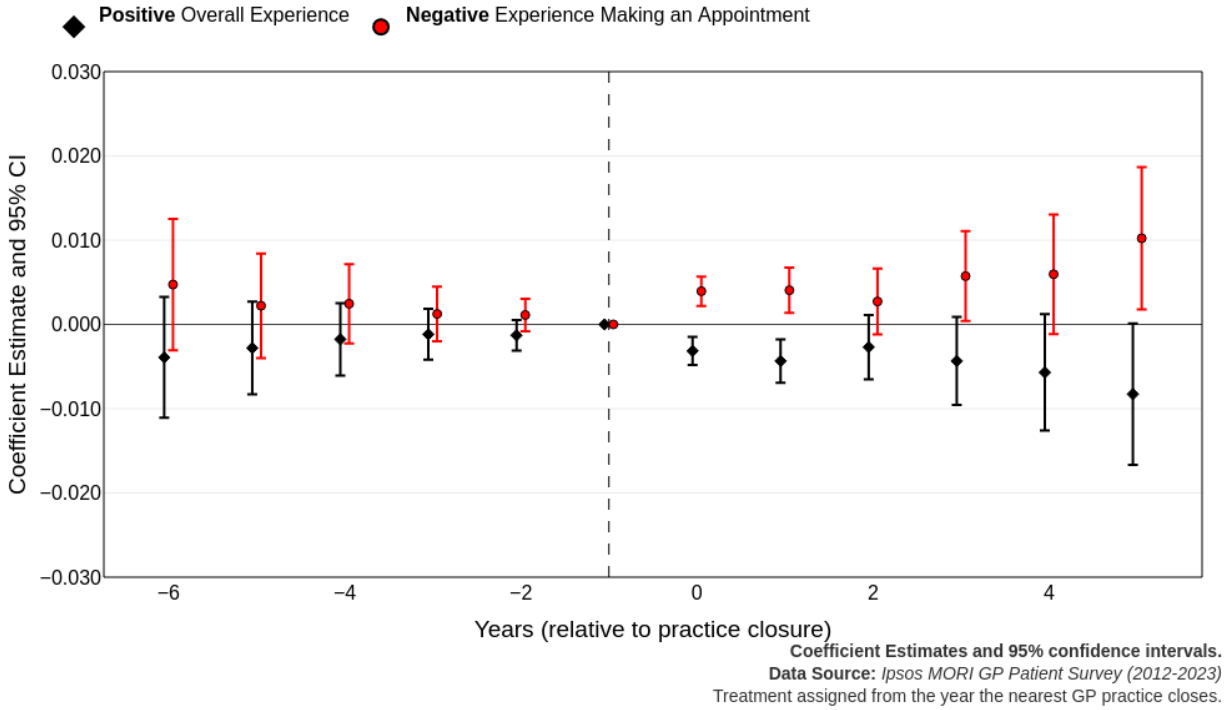
We estimate the following Sun–Abraham event-study model:

$$Y_{it} \;=\; \sum_{g \neq \text{never}} \sum_{\ell \neq -1} \delta_{g\ell} \cdot \mathbf{1}\big[G_i^{\text{nearest}} = g\big] \cdot D_{it}^{\ell} \;+\; \alpha_i \;+\; \alpha_t \;+\; X_{it}'\beta \;+\; \varepsilon_{it},$$

where $D_{it}^{\ell} = \mathbf{1}[t - G_i^{\text{nearest}} = \ell]$ is an indicator for event time ℓ, α_i and α_t denote MSOA and year fixed effects respectively, and X_{it} is the same vector of time-varying controls as in the primary specification. Standard errors are two-way clustered (MSOA $\times$ year).

Figure A6 presents the event-study dynamics for both outcomes using the no-controls specification. Table A5 and Table A6 report the aggregated ATT estimates across four progressively richer model specifications. The results are consistent with the primary MSOA-level analysis: following the closure of the nearest practice, patients at surviving practices report statistically significant deterioration in both overall satisfaction and appointment experience, with effects accumulating over time. This consistency across treatment assignment strategies supports our interpretation that the estimated effects are driven by the patient displacement mechanism—surviving practices absorbing demand from closed neighbours—rather than by the particular geographic granularity used to define treatment.

Figure A6: Effect of Nearest GP Practice Closure on Patient Experiences (Event Study)



Note: Sun–Abraham (2021) event-study estimates for the effect of the nearest GP practice closure on patient satisfaction outcomes. The left panel shows the effect on the share of patients reporting a positive overall experience; the right panel shows the effect on the share reporting a negative experience making an appointment. Treatment is assigned from the year the nearest GP practice closes (`gvar_nearest`). All models include MSOA and year fixed effects. Standard errors are two-way clustered (MSOA × year). The shaded bands are 95% confidence intervals. Period −1 is the omitted reference period.

Table A5: Nearest-Practice Treatment: Effects on Positive Overall Experience (GPPS)

Dependent Variable:	Positive overall experience with GP practice			
Model:	(1)	(2)	(3)	(4)
Variables				
ATT	-0.0047*	-0.0051*	-0.0059**	-0.0025
	(0.0024)	(0.0024)	(0.0026)	(0.0023)
IMD Score		-0.0002	-0.0008*	-0.0010**
		(0.0003)	(0.0004)	(0.0004)
Unemployment Rate			0.0041***	0.0036***
			(0.0010)	(0.0007)
International Migration (inflow)				-0.0060
				(0.0053)
Migrant GP Registrations				0.0035
				(0.0031)
Fixed-effects				
MSOA	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	87,100	85,825	77,926	66,270
R^2	0.61171	0.61433	0.60951	0.54018
Within R^2	0.00207	0.00219	0.00495	0.00559

Clustered (MSOA & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Two-way clustered (MSOA × year) standard errors in parentheses. All models include MSOA and year fixed effects. ATT is the Sun and Abraham (2021) aggregated average treatment effect on the treated. Treatment is assigned from the year in which the nearest GP practice closes (`gvar_nearest`). Model 1: no controls. Model 2: adds IMD Score. Model 3: adds Unemployment Rate. Model 4: adds migration controls (international migration inflow proportion and migrant GP registrations proportion).

Table A6: Nearest-Practice Treatment: Effects on Negative Appointment Experience (GPPS)

Dependent Variable:	Negative overall experience making an appointment			
Model:	(1)	(2)	(3)	(4)
Variables				
ATT	0.0057**	0.0061**	0.0074**	0.0017
	(0.0025)	(0.0025)	(0.0026)	(0.0026)
IMD Score		9.12×10^{-5}	0.0008*	0.0009**
		(0.0004)	(0.0004)	(0.0004)
Unemployment Rate			-0.0049***	-0.0041***
			(0.0012)	(0.0007)
International Migration (inflow)				0.0073
				(0.0053)
Migrant GP Registrations				0.0005
				(0.0038)
Fixed-effects				
MSOA	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	87,027	85,752	77,869	66,266
R^2	0.57859	0.58131	0.57777	0.50710
Within R^2	0.00217	0.00224	0.00543	0.00574

Clustered (MSOA & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Two-way clustered (MSOA × year) standard errors in parentheses. All models include MSOA and year fixed effects. ATT is the Sun and Abraham (2021) aggregated average treatment effect on the treated. Treatment is assigned from the year in which the nearest GP practice closes (`gvar_nearest`). Model 1: no controls. Model 2: adds IMD Score. Model 3: adds Unemployment Rate. Model 4: adds migration controls (international migration inflow proportion and migrant GP registrations proportion).

Understanding Society Analysis

The full results of the analysis of the Understanding Society data are included in this section. The Understanding Society Dataset is a household panel. We use the first two respondents in each household and exclude other members of the household. Under the assumption that the first two respondents are a couple, it is plausible that vote intention within couples is correlated. We therefore additionally provide results with only the *first* respondent in each household below in Figure A8. The results are very similar.

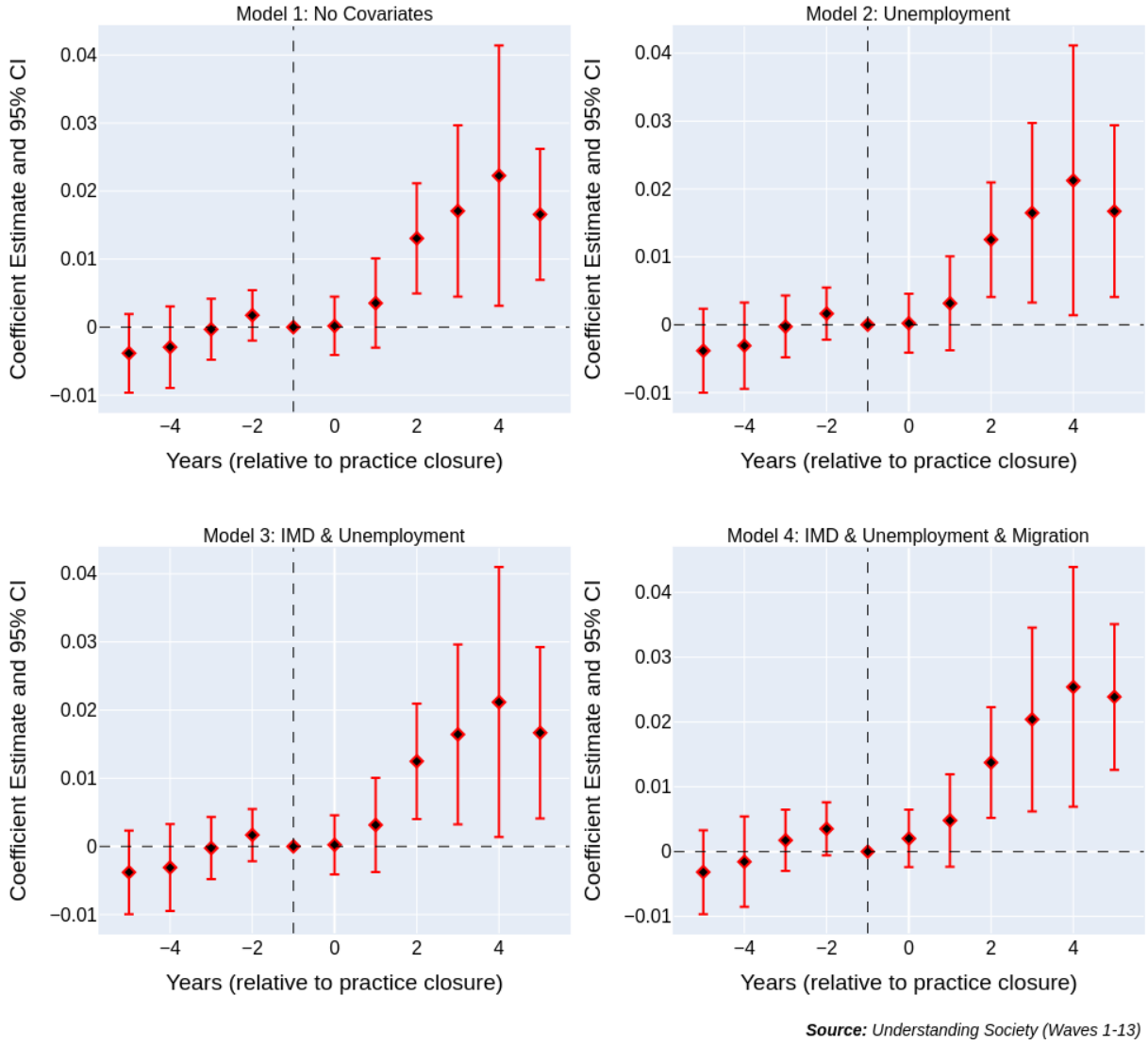
Table A7: Effects of GP Practice Closures on Right-Wing Vote Intentions

	No Controls	IMD Score	Employment Rate + IMD	Migration Controls + All
year = -5	−0.004	−0.004	−0.004	−0.003
	(0.003)	(0.003)	(0.003)	(0.003)
year = -4	−0.003	−0.003	−0.003	−0.002
	(0.003)	(0.003)	(0.003)	(0.004)
year = -3	0.000	0.000	0.000	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
year = -2	0.002	0.002	0.002	0.004
	(0.002)	(0.002)	(0.002)	(0.002)
year = 0	0.000	0.000	0.000	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
year = 1	0.004	0.003	0.003	0.005
	(0.003)	(0.004)	(0.004)	(0.004)
year = 2	0.013***	0.013**	0.012**	0.014**
	(0.004)	(0.004)	(0.004)	(0.004)
year = 3	0.017**	0.016**	0.016**	0.020**
	(0.006)	(0.007)	(0.007)	(0.007)
year = 4	0.022**	0.021*	0.021*	0.025**
	(0.010)	(0.010)	(0.010)	(0.009)
year = 5	0.017***	0.017**	0.017**	0.024***
	(0.005)	(0.006)	(0.006)	(0.006)
IMD Score		0.001	0.001	−0.001
		(0.001)	(0.001)	(0.001)
Employment Rate			0.000	0.000
			(0.000)	(0.000)
Migrant GP Registrations (per pop.)				0.003
				(0.004)
International Migration (per pop.)				0.002
				(0.008)
Num.Obs.	285 156	266 636	266 625	193 710
R2	0.248	0.257	0.257	0.326
R2 Adj.	0.030	0.039	0.039	0.099
R2 Within	0.000	0.000	0.000	0.000
R2 Within Adj.	0.000	0.000	0.000	0.000
AIC	−348 251.7	−310 338.4	−310 316.7	−173 677.5
BIC	328 811.2	322 438.6	322 426.3	323 277.2
RMSE	0.10	0.11	0.11	0.12
FE: ID	X	X	X	X
FE: Year	X	X	X	X
FE: MSOA	X	X	X	X

Clustered (ID & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

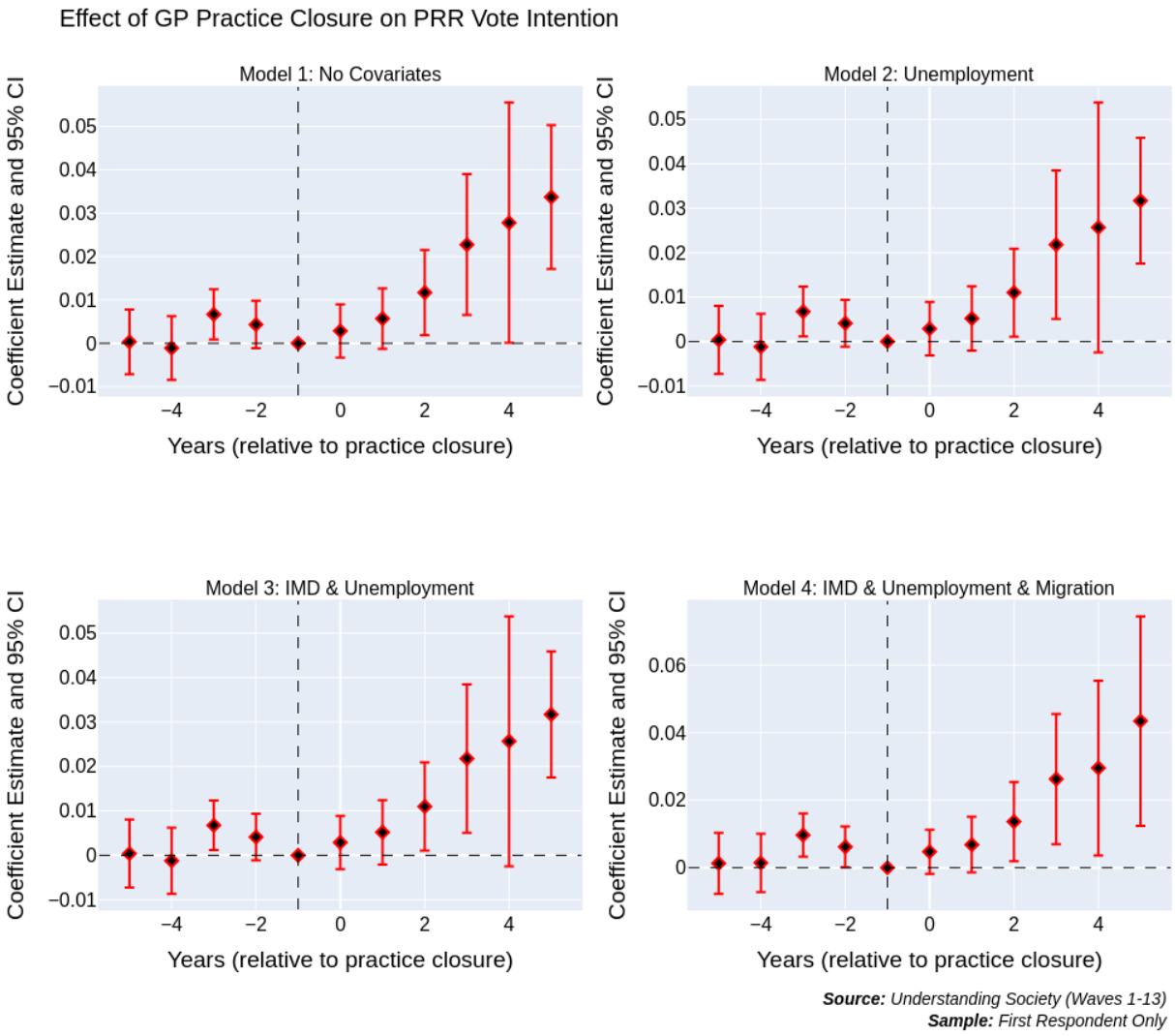
Note: Standard errors clustered at the ID/Year level in parentheses. FEs for respondent, year and MSOA. Model 1 has no control variables. Model 2 conditions on the Index of Multiple Deprivation. Model 3 conditions on unemployment rate and the Index of Multiple Deprivation. Model 4 conditions on IMD, unemployment rate, migrant GP practice registrations and international immigration. There are no estimates for model 4 for year 7 because of missing data in the immigration statistics, which have not been made available from the Office of National Statistics (as of 17 Sept. 2025).

Figure A7: Estimated Effects of GP Practice Closures on Intention to Vote for Populist Right Parties (USOC Data)



Note: Event study estimates and 95% confidence intervals for the effects of GP practice closures on vote intention for PRR parties. Parties include the British National Party, Reform UK, the UK Independence Party and the Brexit Party. Model 1 has no control variables. Model 2 conditions on the Index of Multiple Deprivation. Model 3 conditions on unemployment rate and the Index of Multiple Deprivation. Model 4 conditions on IMD, unemployment rate, migrant GP practice registrations and international immigration.

Figure A8: Estimated Effects of GP Practice Closures on Intention to Vote for Populist Right Parties (USOC Data - First Respondent Only)



Note: Event study estimates and 95% confidence intervals for the effects of GP practice closures on vote intention for PRR parties using only the first respondent in each household. Parties include the British National Party, Reform UK, the UK Independence Party and the Brexit Party. Model 1 has no control variables. Model 2 conditions on the Index of Multiple Deprivation. Model 3 conditions on unemployment rate and the Index of Multiple Deprivation. Model 4 conditions on IMD, unemployment rate, migrant GP practice registrations and international immigration.

Matrix Completion Estimates

Table A8: Dynamic Effects of GP Practice Closures on Vote Intention for Populist Right Party

Time	ATT	S.E.	P-value	Lower Bound (95CI)	Upper Bound (95CI)
-5	-0.000435	0.001100	0.692141	-0.002591	0.001720
-4	0.001182	0.001176	0.314840	-0.001122	0.003485
-3	0.001700	0.001288	0.186898	-0.000825	0.004225
-2	0.001148	0.001210	0.342570	-0.001223	0.003519
-1	0.003109	0.001510	0.039531	0.000149	0.006069
0	0.001777	0.001305	0.173207	-0.000780	0.004335
1	0.010786	0.001934	0.000000	0.006995	0.014577
2	0.005570	0.002308	0.015791	0.001047	0.010094
3	0.006376	0.002560	0.012768	0.001358	0.011394
4	0.008491	0.003269	0.009389	0.002084	0.014898
5	0.007953	0.004415	0.071647	-0.000700	0.016606

Note: Dynamic matrix completion ATT estimates for the effects of practice closures on vote intention for a populist right party. Understanding Society Secure Access Data (University of Essex, Institute for Social and Economic Research 2023). GP practice closures are mapped to voters using the first 5 digits of the postcode of residence (Postcode sector). Estimates are made using the `Fect` library in R (Liu, Wang, and Xu 2022). No control variables are included.

British Election Study Analysis

In this section, the full results of the primary analysis using the British Election Study data to estimate the effects of the closures on vote intention using Equation A1 are presented in table form.

Fixed Effects Estimates

The following results use the Sun and Abraham (2021) cohort-time fixed effects estimator. Fixed effects included for MSOA, Respondent ID and Year. Standard errors clustered by Respondent ID and Year.

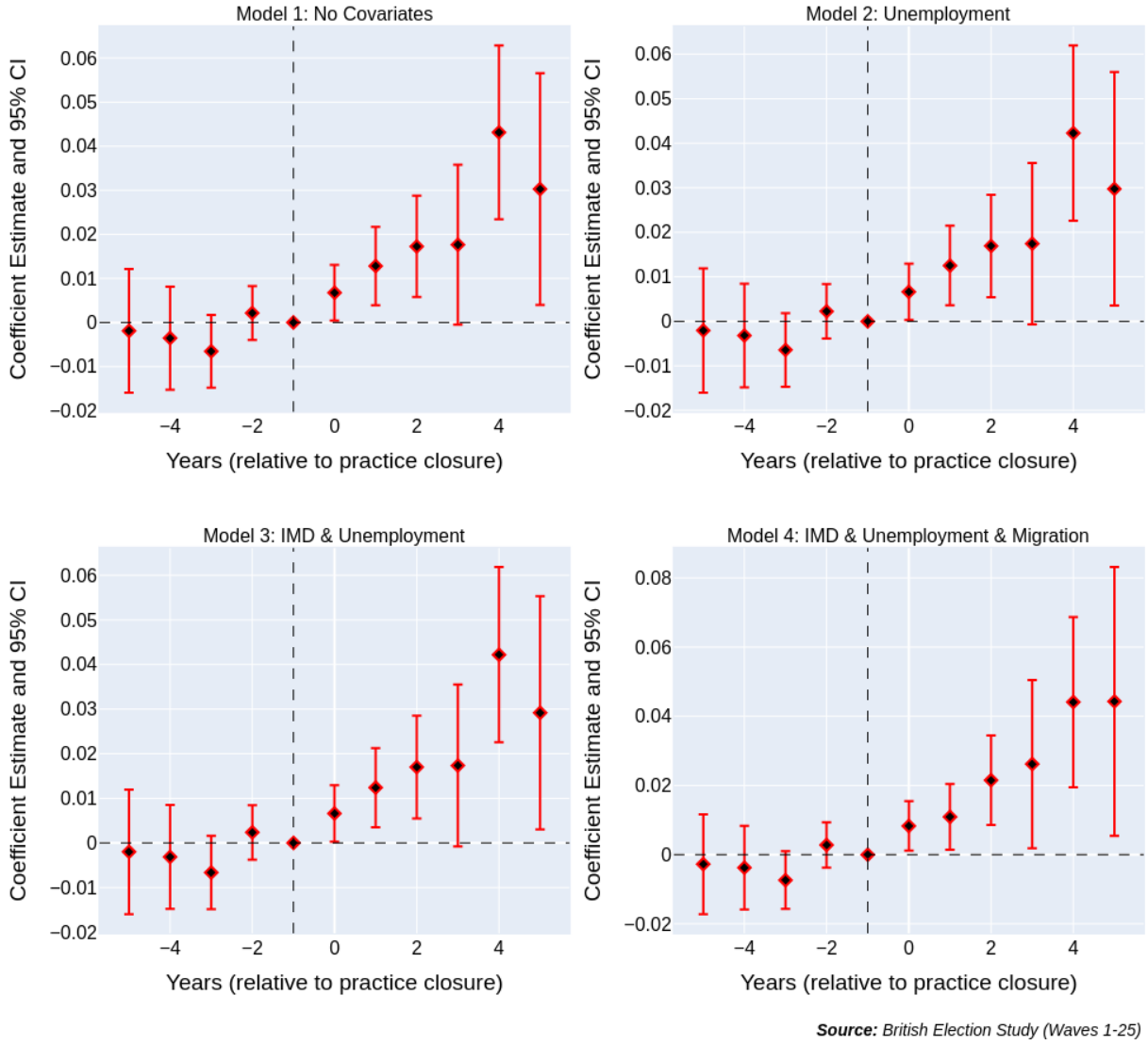
Table A9: Effects of GP Practice Closures on Right-Wing Vote Intentions

	No Controls	IMD Score	Employment Rate + IMD	All controls
year = -5	−0.002	−0.002	−0.002	−0.003
	(0.007)	(0.007)	(0.007)	(0.007)
year = -4	−0.004	−0.003	−0.003	−0.004
	(0.006)	(0.006)	(0.006)	(0.006)
year = -3	−0.007	−0.006	−0.007	−0.007
	(0.004)	(0.004)	(0.004)	(0.004)
year = -2	0.002	0.002	0.002	0.003
	(0.003)	(0.003)	(0.003)	(0.003)
year = 0	0.007*	0.007*	0.007*	0.008*
	(0.003)	(0.003)	(0.003)	(0.004)
year = 1	0.013**	0.013**	0.012**	0.011*
	(0.005)	(0.005)	(0.005)	(0.005)
year = 2	0.017**	0.017**	0.017**	0.021**
	(0.006)	(0.006)	(0.006)	(0.007)
year = 3	0.018*	0.017*	0.017*	0.026*
	(0.009)	(0.009)	(0.009)	(0.012)
year = 4	0.043***	0.042***	0.042***	0.044***
	(0.010)	(0.010)	(0.010)	(0.013)
year = 5	0.030*	0.030*	0.029*	0.044*
	(0.013)	(0.013)	(0.013)	(0.020)
IMD Score		−0.002	−0.002	−0.002
		(0.002)	(0.002)	(0.002)
Employment Rate			0.000*	0.001*
			(0.000)	(0.000)
International Migration (per pop.)				−1.232
				(1.094)
Migrant GP Registrations (per pop.)				2.081**
				(0.731)
Num.Obs.	497 860	497 391	497 252	466 386
R2	0.578	0.579	0.579	0.587
R2 Adj.	0.488	0.488	0.488	0.494
R2 Within	0.000	0.000	0.000	0.000
R2 Within Adj.	0.000	0.000	0.000	0.000
AIC	826.2	715.2	809.3	12 963.4
BIC	978 282.8	977 255.1	977 057.9	963 004.2
RMSE	0.20	0.20	0.20	0.20
Std.Errors	by: ID & year	by: ID & year	by: ID & year	by: ID & year
FE: ID	X	X	X	X
FE: Year	X	X	X	X
FE: MSOA	X	X	X	X

Clustered (ID & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Standard errors clustered at the ID/Year level in parentheses. Standard errors clustered by ID/year. Model 1 has no control variables. Model 2 conditions on the Index of Multiple Deprivation. Model 3 conditions on unemployment rate and the Index of Multiple Deprivation. Model 4 conditions on IMD, unemployment rate, migrant GP practice registrations and international immigration. There are no estimates for model 4 for year 7 because of missing data in the immigration statistics, which have not been made available from the Office of National Statistics (as of 17 Sept. 2025).

Figure A9: Estimated Effects of GP Practice Closures on Intention to Vote for Populist Right Parties (BES Data)



Note: Event study estimates and 95% confidence intervals for the effects of GP practice closures on vote intention for PRR parties. Parties include the British National Party, Reform UK, the UK Independence Party and the Brexit Party. Models 1-4 correspond to the models in Table A9. Model 1 has no control variables. Model 2 conditions on the Index of Multiple Deprivation. Model 3 conditions on unemployment rate and the Index of Multiple Deprivation. Model 4 conditions on IMD, unemployment rate, migrant GP practice registrations and international immigration. There are no estimates for model 4 for year 7 because of missing data in the immigration statistics, which have not been made available from the Office of National Statistics (as of 17 Sept. 2025).

Matrix Completion Estimates

Table A10: Dynamic Effects of Closures on Vote Intention for PRR Parties

Time (Wave)	ATT	S.E.	p.value	CI.lower	CI.upper
-11	-0.001883	0.004809	0.695440	-0.011307	0.007542
-10	0.001911	0.005744	0.739408	-0.009348	0.013169
-9	-0.006622	0.004258	0.119894	-0.014966	0.001723
-8	0.001037	0.004234	0.806455	-0.007261	0.009335
-7	-0.005456	0.004031	0.175823	-0.013356	0.002444
-6	-0.000773	0.004122	0.851222	-0.008851	0.007305
-5	-0.003222	0.004421	0.466054	-0.011887	0.005442
-4	-0.000647	0.004154	0.876147	-0.008789	0.007494
-3	0.006449	0.004421	0.144574	-0.002215	0.015113
-2	0.001639	0.004187	0.695396	-0.006567	0.009845
-1	0.002785	0.004254	0.512767	-0.005554	0.011123
0	0.009224	0.004190	0.027695	0.001012	0.017435
1	0.021235	0.008299	0.010504	0.004969	0.037500
2	0.012669	0.010839	0.242479	-0.008575	0.033912
3	0.016233	0.010724	0.130101	-0.004786	0.037252
4	0.039731	0.012189	0.001116	0.015841	0.063620
5	0.035219	0.011092	0.001498	0.013479	0.056960
6	0.023321	0.011284	0.038751	0.001206	0.045437
7	0.037053	0.015461	0.016548	0.006751	0.067356
8	0.040310	0.016227	0.012985	0.008506	0.072114
9	0.018183	0.014020	0.194649	-0.009295	0.045662
10	0.037228	0.016264	0.022079	0.005352	0.069105
11	0.025474	0.017291	0.140688	-0.008416	0.059364

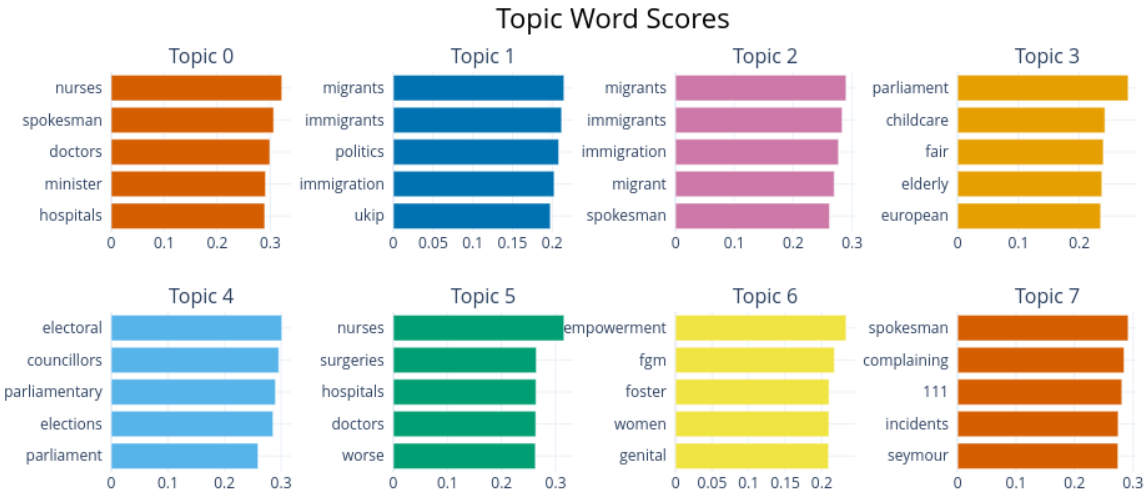
Note: Dynamic matrix completion ATT estimates for the effects of practice closures on vote intention for the British National Party, UKIP, Reform UK and the Brexit Party. Estimates are made using the `Fect` library in R (Liu, Wang, and Xu 2022). No control variables are included. Fixed effects for ID/Wave. Models use the survey wave rather than survey year for fixed effects to accommodate the data requirements for matrix completion (e.g. uniquely indexed data).

Topic Model

We consider the ways in which the United Kingdom Independence Party frames the NHS by creating topic models of official press releases from the party that address the topic directly. We rely on `BerTopic` to estimate the models (Grootendorst 2022) and use data from Dickson and Hobolt (2024).

We first narrow the data to include only press releases that contain public healthcare related terminology using the lexicon developed in Table A32. We then estimate a topic model using the 237 press releases that were identified with the lexicon. We present the keywords associated with each of the topics identified by the model in Figure A10.

Figure A10: UKIP Framing of the NHS and GP Services - Top keywords



Note: Topic model includes all official press releases from UKIP that make mention of the NHS or GPs. Press release data includes 2010-2022. Data are from Dickson and Hobolt (2024).

Moderation Analysis

We provide the full results in table format from the moderation analysis conducted in the main text.

Populist Radical Right

CATT estimates for the conditional effects of the closures on PRR party vote intention.

Table A11: Moderator: Index of Multiple Deprivation Score (PRR)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.000	0.010	-0.000	1.000	-0.019	0.019	Index of Multiple Deprivation Score
75% vs 50%	-0.000	0.010	-0.000	1.000	-0.019	0.019	Index of Multiple Deprivation Score
75% vs 25%	-0.000	0.020	-0.000	1.000	-0.039	0.039	Index of Multiple Deprivation Score

Table A12: Moderator: Employment Rate (PRR)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.007	0.008	-0.849	0.396	-0.022	0.009	Employment Rate
75% vs 50%	-0.007	0.008	-0.849	0.396	-0.022	0.009	Employment Rate
75% vs 25%	-0.014	0.016	-0.849	0.396	-0.045	0.018	Employment Rate

Table A13: Moderator: (log) International Migration Per Capita (PRR)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.016	0.007	2.358	0.018	0.003	0.029	(log) International Migration Per Capita
75% vs 50%	0.016	0.007	2.358	0.018	0.003	0.029	(log) International Migration Per Capita
75% vs 25%	0.032	0.014	2.358	0.018	0.005	0.059	(log) International Migration Per Capita

Table A14: Moderator: (log) Migrant GP Registrations Per Capita (PRR)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.015	0.008	2.033	0.042	0.001	0.030	(log) Migrant GP Registrations Per Capita
75% vs 50%	0.015	0.008	2.033	0.042	0.001	0.030	(log) Migrant GP Registrations Per Capita
75% vs 25%	0.031	0.015	2.033	0.042	0.001	0.060	(log) Migrant GP Registrations Per Capita

Labour Party

CATT estimates for the conditional effects of the closures on Labour Party vote intention.

Table A15: Moderator: (log) Employment Rate (Labour)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.006	0.008	0.764	0.445	-0.010	0.022	(log) Employment Rate
75% vs 50%	0.006	0.008	0.764	0.445	-0.010	0.022	(log) Employment Rate
75% vs 25%	0.012	0.016	0.764	0.445	-0.019	0.044	(log) Employment Rate

Table A16: Moderator: Index of Multiple Deprivation Score (Labour)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.009	0.011	-0.842	0.400	-0.030	0.012	Index of Multiple Deprivation Score
75% vs 50%	-0.009	0.011	-0.842	0.400	-0.030	0.012	Index of Multiple Deprivation Score
75% vs 25%	-0.018	0.022	-0.842	0.400	-0.061	0.024	Index of Multiple Deprivation Score

Table A17: Moderator: (log) International Migration Per Capita (Labour)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.003	0.008	0.353	0.724	-0.014	0.020	(log) International Migration Per Capita
75% vs 50%	0.003	0.008	0.353	0.724	-0.014	0.020	(log) International Migration Per Capita
75% vs 25%	0.006	0.017	0.353	0.724	-0.027	0.039	(log) International Migration Per Capita

Table A18: Moderator: (log) Migrant GP Registrations Per Capita (Labour)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.003	0.009	0.360	0.719	-0.015	0.022	(log) Migrant GP Registrations Per Capita
75% vs 50%	0.003	0.009	0.360	0.719	-0.015	0.022	(log) Migrant GP Registrations Per Capita
75% vs 25%	0.007	0.019	0.360	0.719	-0.030	0.044	(log) Migrant GP Registrations Per Capita

Green Party

CATT estimates for the conditional effects of the closures on Green Party vote intention.

Table A19: Moderator: (log) Employment Rate (Green Party)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.002	0.005	-0.327	0.744	-0.011	0.008	(log) Employment Rate
75% vs 50%	-0.002	0.005	-0.327	0.744	-0.011	0.008	(log) Employment Rate
75% vs 25%	-0.003	0.009	-0.327	0.744	-0.021	0.015	(log) Employment Rate

Table A20: Moderator: Index of Multiple Deprivation Score (Green Party)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.002	0.005	0.399	0.690	-0.009	0.013	Index of Multiple Deprivation Score
75% vs 50%	0.002	0.005	0.399	0.690	-0.009	0.013	Index of Multiple Deprivation Score
75% vs 25%	0.004	0.011	0.399	0.690	-0.017	0.026	Index of Multiple Deprivation Score

Table A21: Moderator: (log) International Migration Per Capita (Green Party)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.007	0.005	1.392	0.164	-0.003	0.017	(log) International Migration Per Capita
75% vs 50%	0.007	0.005	1.392	0.164	-0.003	0.017	(log) International Migration Per Capita
75% vs 25%	0.014	0.010	1.392	0.164	-0.006	0.033	(log) International Migration Per Capita

Table A22: Moderator: (log) Migrant GP Registrations Per Capita (Green Party)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.009	0.005	1.667	0.095	-0.002	0.020	(log) Migrant GP Registrations Per Capita
75% vs 50%	0.009	0.005	1.667	0.095	-0.002	0.020	(log) Migrant GP Registrations Per Capita
75% vs 25%	0.018	0.011	1.667	0.095	-0.003	0.039	(log) Migrant GP Registrations Per Capita

Liberal Democrats

CATT estimates for the conditional effects of the closures on Liberal Democrat vote intention.

Table A23: Moderator: (log) Employment Rate (LibDem)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.017	0.006	2.744	0.006	0.005	0.030	(log) Employment Rate
75% vs 50%	0.017	0.006	2.744	0.006	0.005	0.030	(log) Employment Rate
75% vs 25%	0.035	0.013	2.744	0.006	0.010	0.059	(log) Employment Rate

Table A24: Moderator: Index of Multiple Deprivation Score (LibDem)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.013	0.007	-1.750	0.080	-0.027	0.002	Index of Multiple Deprivation Score
75% vs 50%	-0.013	0.007	-1.750	0.080	-0.027	0.002	Index of Multiple Deprivation Score
75% vs 25%	-0.025	0.015	-1.750	0.080	-0.054	0.003	Index of Multiple Deprivation Score

Table A25: Moderator: (log) International Migration Per Capita (LibDem)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.009	0.006	-1.587	0.113	-0.020	0.002	(log) International Migration Per Capita
75% vs 50%	-0.009	0.006	-1.587	0.113	-0.020	0.002	(log) International Migration Per Capita
75% vs 25%	-0.018	0.011	-1.587	0.113	-0.040	0.004	(log) International Migration Per Capita

Table A26: Moderator: (log) Migrant GP Registrations Per Capita (LibDem)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.008	0.006	-1.303	0.193	-0.020	0.004	(log) Migrant GP Registrations Per Capita
75% vs 50%	-0.008	0.006	-1.303	0.193	-0.020	0.004	(log) Migrant GP Registrations Per Capita
75% vs 25%	-0.016	0.012	-1.303	0.193	-0.041	0.008	(log) Migrant GP Registrations Per Capita

Conservative Party

CATT estimates for the conditional effects of the closures on Conservative Party vote intention.

Table A27: Moderator: (log) Employment Rate (Conservatives)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.005	0.010	-0.493	0.622	-0.023	0.014	(log) Employment Rate
75% vs 50%	-0.005	0.010	-0.493	0.622	-0.023	0.014	(log) Employment Rate
75% vs 25%	-0.009	0.019	-0.493	0.622	-0.047	0.028	(log) Employment Rate

Table A28: Moderator: Index of Multiple Deprivation Score (Conservatives)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	0.009	0.012	0.743	0.457	-0.014	0.032	Index of Multiple Deprivation Score
75% vs 50%	0.009	0.012	0.743	0.457	-0.014	0.032	Index of Multiple Deprivation Score
75% vs 25%	0.018	0.024	0.743	0.457	-0.029	0.064	Index of Multiple Deprivation Score

Table A29: Moderator: (log) International Migration Per Capita (Conservatives)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.024	0.008	-2.922	0.003	-0.040	-0.008	(log) International Migration Per Capita
75% vs 50%	-0.024	0.008	-2.922	0.003	-0.040	-0.008	(log) International Migration Per Capita
75% vs 25%	-0.047	0.016	-2.922	0.003	-0.079	-0.016	(log) International Migration Per Capita

Table A30: Moderator: (log) Migrant GP Registrations Per Capita (Conservatives)

	Estimate	Std. Error	t-value	p-value	2.5% CI	97.5% CI	Moderator
50% vs 25%	-0.026	0.009	-2.855	0.004	-0.044	-0.008	(log) Migrant GP Registrations Per Capita
75% vs 50%	-0.026	0.009	-2.855	0.004	-0.044	-0.008	(log) Migrant GP Registrations Per Capita
75% vs 25%	-0.053	0.018	-2.855	0.004	-0.089	-0.017	(log) Migrant GP Registrations Per Capita

(Re-)Opening Practices Robustness Check

In the primary specifications, we drop from the analysis all individuals who are exposed to a closure but who later are also exposed to a newly opened practice. To ensure our results are robust to this decision, we re-estimate the main models while considering these individuals as treated (model 1) and when considering them as controls (model 2). We estimate effects using our primary specification in Equation A1. The results are similar to the results presented in the text.

Table A31: Event Study Robustness Check with BES: Different Configurations of Open After Close

Dependent Variable:	PRR Vote Intention	
Model:	(1)	(2)
Variables		
year = -5	−0.005	−0.005
	(0.007)	(0.007)
year = -4	−0.007	−0.007
	(0.005)	(0.005)
year = -3	−0.008*	−0.008*
	(0.004)	(0.004)
year = -2	0.001	0.001
	(0.003)	(0.003)
year = 0	0.007*	0.007*
	(0.003)	(0.003)
year = 1	0.014***	0.014***
	(0.004)	(0.004)
year = 2	0.018***	0.018***
	(0.005)	(0.005)
year = 3	0.018*	0.018*
	(0.008)	(0.008)
year = 4	0.037***	0.037***
	(0.008)	(0.008)
year = 5	0.029**	0.029**
	(0.010)	(0.010)
Fixed-effects		
Respondent ID	Yes	Yes
Year	Yes	Yes
MSOA11	Yes	Yes
Fit statistics		
Observations	518,092	518,092
R^2	0.57486	0.57486
Within R^2	0.00026	0.00026

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Standard errors clustered at the ID/Year level in parentheses. This analysis is part of the Robustness checks for the event study design, testing different configurations of the 'open after close' treatment assignment.

Linking the NHS with Immigration in Party Communication

This section documents methods used to measure the extent to which UK populist-right parties connect NHS decline to immigration in their official communications. We analyze two corpora: (i) *YouTube transcripts* from official UKIP and Reform UK channels (2011–mid-2024), and (ii) *UKIP press releases* (2010–2024). The design harmonizes definitions across corpora while respecting their different textual structures (punctuation and sentence boundaries in press releases vs. unreliable punctuation in YouTube transcripts).

Dictionaries and connectors

We use three case-insensitive regex lexicons (see Table A32 for full lists):

- L_E (**NHS / public-service terms**): e.g., `NHS`, `GP(s)`, `A&E`, `waiting list/time`, `hospital(s)`, `nurse shortage`, etc.
- L_N (**Immigration terms**): e.g., `immigration/immigrant(s)/migrant(s)`, `asylum seeker(s)`, `small boats`, `mass immigration`, etc.
- L_C (**causal / burden connectors**): e.g., `because`, `due to`, `leads to`, `results in`, `puts pressure on`, `affects the NHS`, `overwhelms`, etc.

We report two complementary quantities. The first – *co-mention* –captures simple co-location of NHS and immigration language. The second –*close-proximity causal linkage* – asks for more: in the same local span, we also require a causal/burden connector. Intuitively, the first tells us "how often they talk about the NHS and immigration together"; the second tells us "how often they frame immigration as causing pressure on the NHS".

YouTube transcripts: windowing and measures

Transcripts often lack reliable punctuation. We therefore segment each video into overlapping token windows of length w=50 with step s=10 (a "soft sentence"), and we treat δ=12 tokens as the proximity threshold for connectors.

Let W denote a window, and define

$$I_E(W) = \mathbf{1}\{W \cap L_E \neq \emptyset\}, \qquad I_N(W) = \mathbf{1}\{W \cap L_N \neq \emptyset\}.$$

For connectors,

$$I_C(W;\delta) = \mathbf{1}\{\exists c \in L_C \text{ within } \pm\,\delta \text{ tokens of any } L_E \text{ or } L_N \text{ term in } W\}.$$

Let $\mathcal{W}_m$ be all windows from videos uploaded in month m. We compute two monthly rates:

$$r_m^{\text{cooc}} = \frac{1}{|\mathcal{W}_m|} \sum_{W \in \mathcal{W}_m} \mathbf{1}\{I_E(W) = 1 \wedge I_N(W) = 1\},$$

$$r_m^{\text{link}} = \frac{1}{|\mathcal{W}_m|} \sum_{W \in \mathcal{W}_m} \mathbf{1}\{I_E(W) = 1 \wedge I_N(W) = 1 \wedge I_C(W; \delta) = 1\}.$$

The first rate reflects thematic overlap; the second reflects a nearby causal/burden formulation (e.g., "immigration *puts pressure on* the NHS"). For presentation, we plot centered 3-month moving averages and shade months with low text volume.

UKIP press releases: sentence-based design

Press releases have reliable sentence boundaries, so we operate at the sentence level and allow links expressed across adjacent sentences.

Sentence-level indicators. For sentence s in document d, define

$$I_E(s) = \mathbf{1}\{s \cap L_E \neq \emptyset\}, \qquad I_N(s) = \mathbf{1}\{s \cap L_N \neq \emptyset\}, \qquad I_C(s) = \mathbf{1}\{s \cap L_C \neq \emptyset\}.$$

We register (i) **same-sentence** co-mentions and linkages, and (ii) **adjacent-sentence** linkages where immigration+connector in s directly neighbors an NHS sentence in $s-1$ or $s+1$ (or vice versa):

$$\begin{aligned}
\text{Same-sentence co-mention}: &\ I_E(s) \wedge I_N(s), \\
\text{Same-sentence linkage}: &\ I_E(s) \wedge I_N(s) \wedge I_C(s), \\
\text{Adjacent-sentence linkage}: &\ \big(I_N(s) \wedge I_C(s)\big) \wedge \big(I_E(s-1) \vee I_E(s+1)\big) \text{ or the reverse.}
\end{aligned}$$

Document-level labels and aggregation. A document is flagged as a co-mention if any sentence contains both L_E and L_N. It is flagged as a causal linkage if either a same-sentence linkage or an adjacent-sentence linkage occurs. Let D_m denote documents released in month m; the monthly rates are

$$r_{m,\text{press}}^{\text{cooc}} = \frac{1}{|D_m|} \sum_{d \in D_m} \mathbf{1}\{\text{co-mention in } d\},$$

$$r_{m,\text{press}}^{\text{link}} = \frac{1}{|D_m|} \sum_{d \in D_m} \mathbf{1}\{\text{same- or adjacent-sentence linkage in } d\}.$$

We plot centered 3-month moving averages and shade low-volume months in the figures in the main text.

Interpretation

Co-mention captures topical coupling (the NHS and immigration appear together). *Close-proximity causal linkage* requires that coupling *plus* an explicit causal or burden connector in the same local span. In practice, this distinction separates passages that merely juxtapose topics ("NHS and immigration") from those that attribute NHS strain to immigration ("immigration *leads to* longer NHS waits"). We favor short windows and a modest proximity threshold to approximate a clause-level claim in noisy transcripts, while the press-release design respects sentences and allows immediately adjacent causal constructions. Connector coverage is documented in Table A32.

Table A32: Lexicons for Public Services–Immigration Linkage Analysis

Public Services Terms	**Immigration Terms**
nhs; national health; national healthcare; healthcare; health service; national health service; public health; public healthcare; healthcare system; gp; general practitioner; family doctor; doctor shortage; nurse shortage; hospital; hospitals; accident and emergency; a&e; waiting list; waiting lists; waiting time; waiting times	immigrant; immigrants; immigration; migrant; migrants; migrant workers; illegal immigration; illegal immigrant; illegal immigrants; asylum seeker; asylum seekers; refugee; refugees; border control; border security; channel crossing; channel crossings; small boat; small boats; deportation; migrant crisis; foreigner; foreigners; overcrowding; overcrowded; population growth; too many people; too much immigration; mass immigration

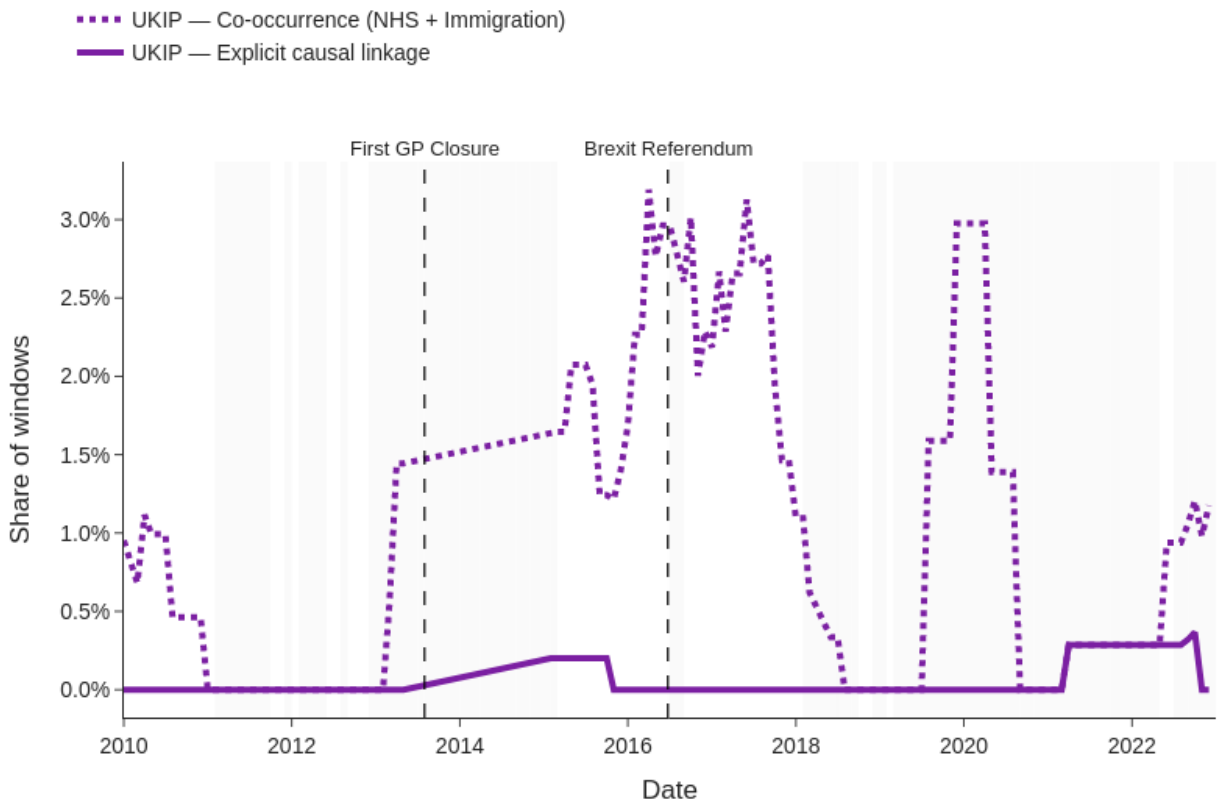
Table A33: Causal connectors for text analysis lexicon. Patterns are Python-style regex, matched case-insensitively

Regex pattern
because
due to
thanks to
as a result of
caus(?:e|ed|es|ing)\b
driv(?:e|en|es|ing)\b
lead(?:s|ing)? to
result(?:s|ing)? in
put(?:s|ting)? pressure on
put(?:s|ting)? strain on
push(?:es|ing)? .* over the edge
overwhelm(?:s|ing)?
overburden(?:s|ing)?
contribut(?:e|es|ed|ing)\b to
linked to
responsible for
impact(?:s|ed|ing)? (?:on)?(?:the)?
harm(?:s|ed|ing)? (?:the)?
damage(?:s|ed|ing)? (?:the)?
undermine(?:s|d|ing)? (?:the)?
crowd(?:s|ed|ing)? (?:the)?
increas(?:e|es|ed|ing)\b (?:the)?(?:burden|pressure|strain|demand)
worsen(?:s|ed|ing)? (?:the)?(?:crisis|situation|problems?)
exacerbat(?:e|es|ed|ing)\b (?:the)?(?:crisis|situation|problems?)
add(?:s|ed|ing)? to (?:the)?(?:burden|pressure|strain|demand)
fuel(?:s|ed|ing)? (?:the|our)?(?:crisis|problems?)
stretch(?:es|ed|ing)? (?:the|our)?
tax(?:es|ed|ing)? (?:the|our)?
overload(?:s|ed|ing)? (?:the|our)?
overstretch(?:es|ed|ing)? (?:the|our)?
affect(?:s|ed|ing)? (?:the|our)?
swamp(?:s|ed|ing)? (?:the|our)?
clog(?:s|ged|ging)? (?:the|our)?
harm(?:s|ed|ing)? (?:the|our)?
damage(?:s|ed|ing)? (?:the|our)?
undermine(?:s|d|ing)? (?:the|our)?
crowd(?:s|ed|ing)? (?:our)?
increas(?:e|es|ed|ing)\b (?:our)?(?:burden|pressure|strain|demand)
worsen(?:s|ed|ing)? (?:our)?(?:crisis|situation|problems?)
exacerbat(?:e|es|ed|ing)\b (?:our)?(?:crisis|situation|problems?)
add(?:s|ed|ing)? to (?:our)?(?:burden|pressure|strain|demand)
fuel(?:s|ed|ing)? (?:our)?(?:crisis|problems?)
stretch(?:es|ed|ing)? (?:our)?
tax(?:es|ed|ing)? (?:our)?
overload(?:s|ed|ing)? (?:our)?
overstretch(?:es|ed|ing)? (?:our)?
affect(?:s|ed|ing)? (?:our)?
swamp(?:s|ed|ing)? (?:our)?
clog(?:s|ged|ging)? (?:our)?
rais(?:e|es|ed|ing)\b demand
make(?:s|ing|d)? .* (worse|collapse|break down)
keep(?:s|ing)? under strain
blame
is why .* (nhs|hospital|public service)
means .* (nhs|hospital|public service)

Note: The connectors are expressed as regex patterns (Python syntax); matching is case-insensitive.

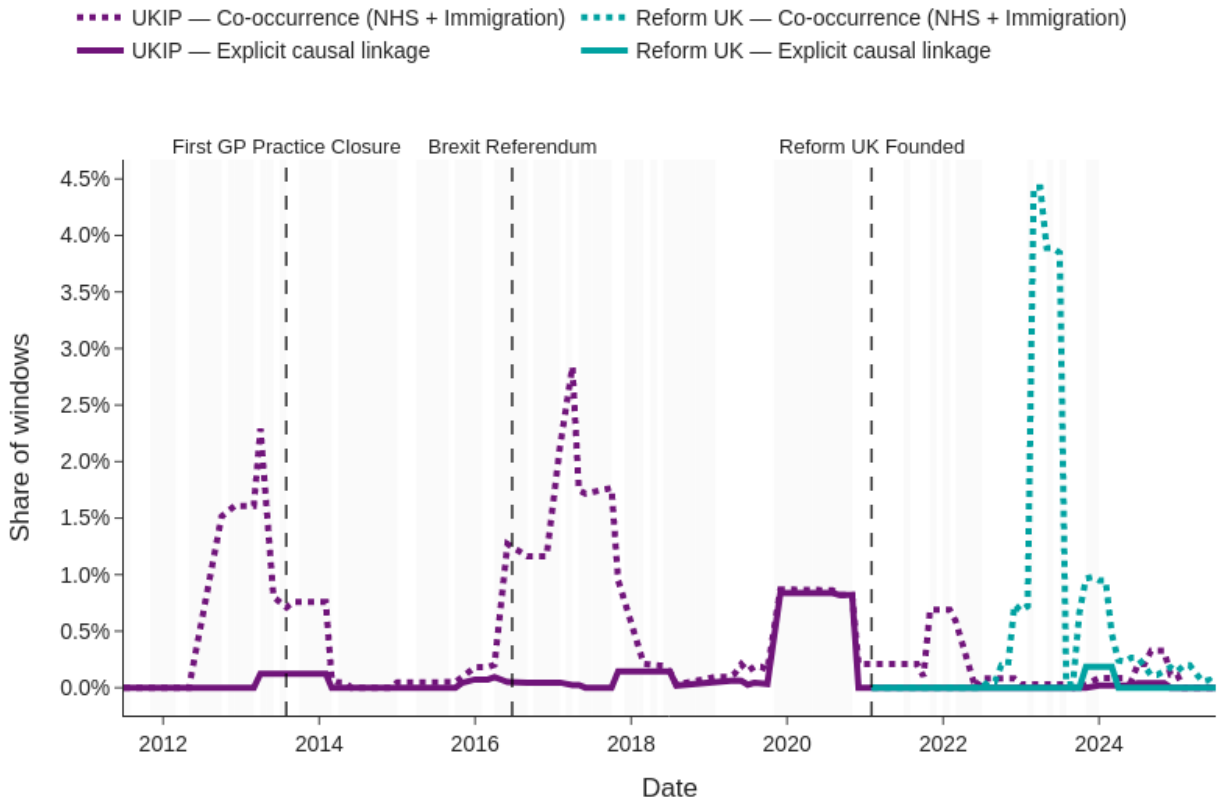
Extended Results

Figure A11: Linking NHS Decline to Migration in UKIP Press Releases



Note: Topical co-occurence with causal linkage words in UKIP press releases. Figure uses a 20-window moving average. Periods of greater uncertainty (e.g. fewer press releases) identified in gray. Full methodology in section .

Figure A12: Linking NHS Decline to Migration in YouTube Videos

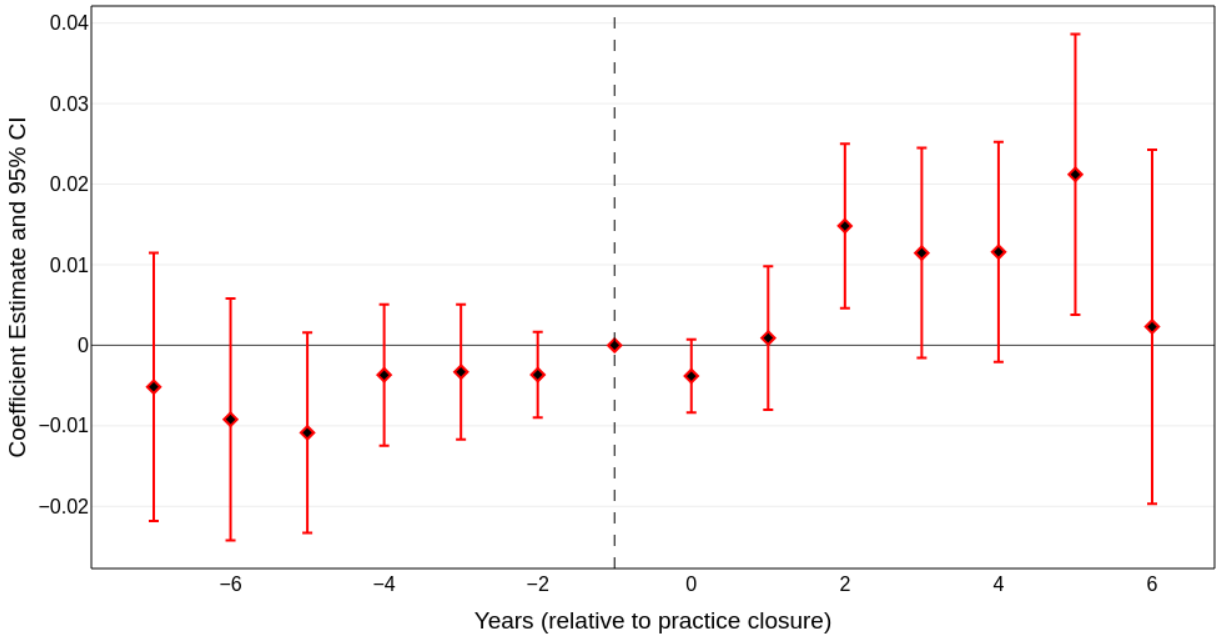


Note: Topical co-occurence with causal linkage words in UKIP and Reform UK YouTube Videos. Figure presents the share of windows ($w = 50$ tokens) in each month that contains (i) co-occurrence; (ii) co-occurrence with proximate causal connectors. Figure uses a 5-window moving average. Periods of greater uncertainty (e.g. fewer YouTube videos) are identified in gray. Full methodology in section .

Placebo Test: Reported *Past* Vote

Figure A13 presents the effects of GP practice closures on reported past vote in the most recent previous election. The results suggest that about 1-2 percent of individuals who had their GP practice close started reporting that they voted for a PRR party in the past election approximately two years after the closure. We estimate effects using our primary specification in Equation A1.

Figure A13: Effects of GP Practice Closure on Reported Past Vote



Source: *British Election Study MSOA Data (Waves 1-25)*
DV: *Past Reported Vote Intention for PRR Party (UKIP, BNP, Reform, Brexit Party)*

Note: Placebo estimates for the effects of a GP practice closure on reported vote in the most recent past election. No control variables are included. Fixed effects included for individuals, years, and MSOA, with standard errors clustered by year and ID.

Robustness Check: Not-yet-treated as the Control Group

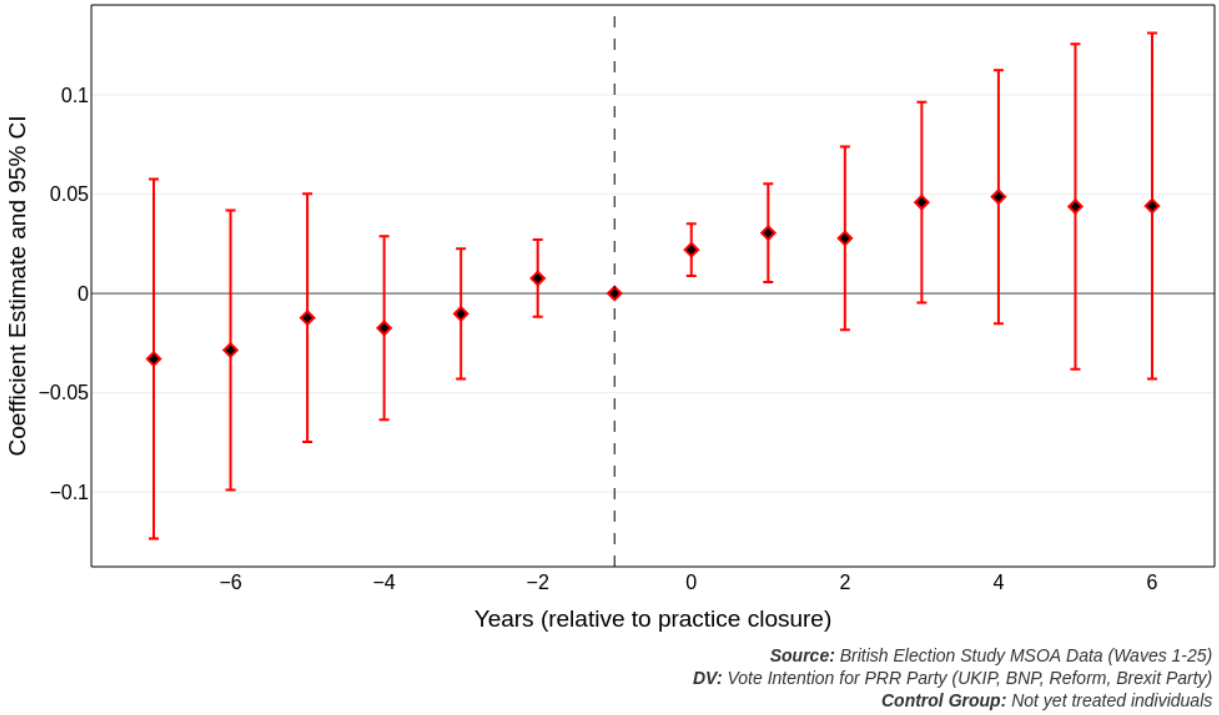
The section includes a robustness check in which we subset the data to include only treated and not-yet-treated individuals. We then estimate the effects of the closures using as the control group individuals that will eventually be exposed to a closure. This process leaves 66,467 observations. We estimate effects using our primary specification in Equation A1.

Table A34: Effects of GP Practice Closure with Not-yet-treated as Control Group

Covariate	Estimate	Std. Error	t value	$Pr(>\|t\|)$	outcome
year: -5	-0.012288	0.031896	-0.385250	0.708999	PRR vote intention (control: not yet treated)
year: -4	-0.017450	0.023562	-0.740585	0.477821	PRR vote intention (control: not yet treated)
year: -3	-0.010251	0.016696	-0.613965	0.554438	PRR vote intention (control: not yet treated)
year: -2	0.007597	0.009902	0.767263	0.462578	PRR vote intention (control: not yet treated)
year: 0	0.021976	0.006660	3.299620	0.009235	PRR vote intention (control: not yet treated)
year: 1	0.030452	0.012616	2.413803	0.039005	PRR vote intention (control: not yet treated)
year: 2	0.027779	0.023515	1.181325	0.267748	PRR vote intention (control: not yet treated)
year: 3	0.045900	0.025784	1.780139	0.108752	PRR vote intention (control: not yet treated)
year: 4	0.048667	0.032547	1.495281	0.169054	PRR vote intention (control: not yet treated)
year: 5	0.043792	0.041782	1.048124	0.321912	PRR vote intention (control: not yet treated)

Note: Estimates and 95% confidence intervals. Standard errors clustered by ID/year. No control variables are included. Fixed effects included for individuals, years, and MSOAs.

Figure A14: Effects of GP Practice Closure (Not-yet-treated as control)

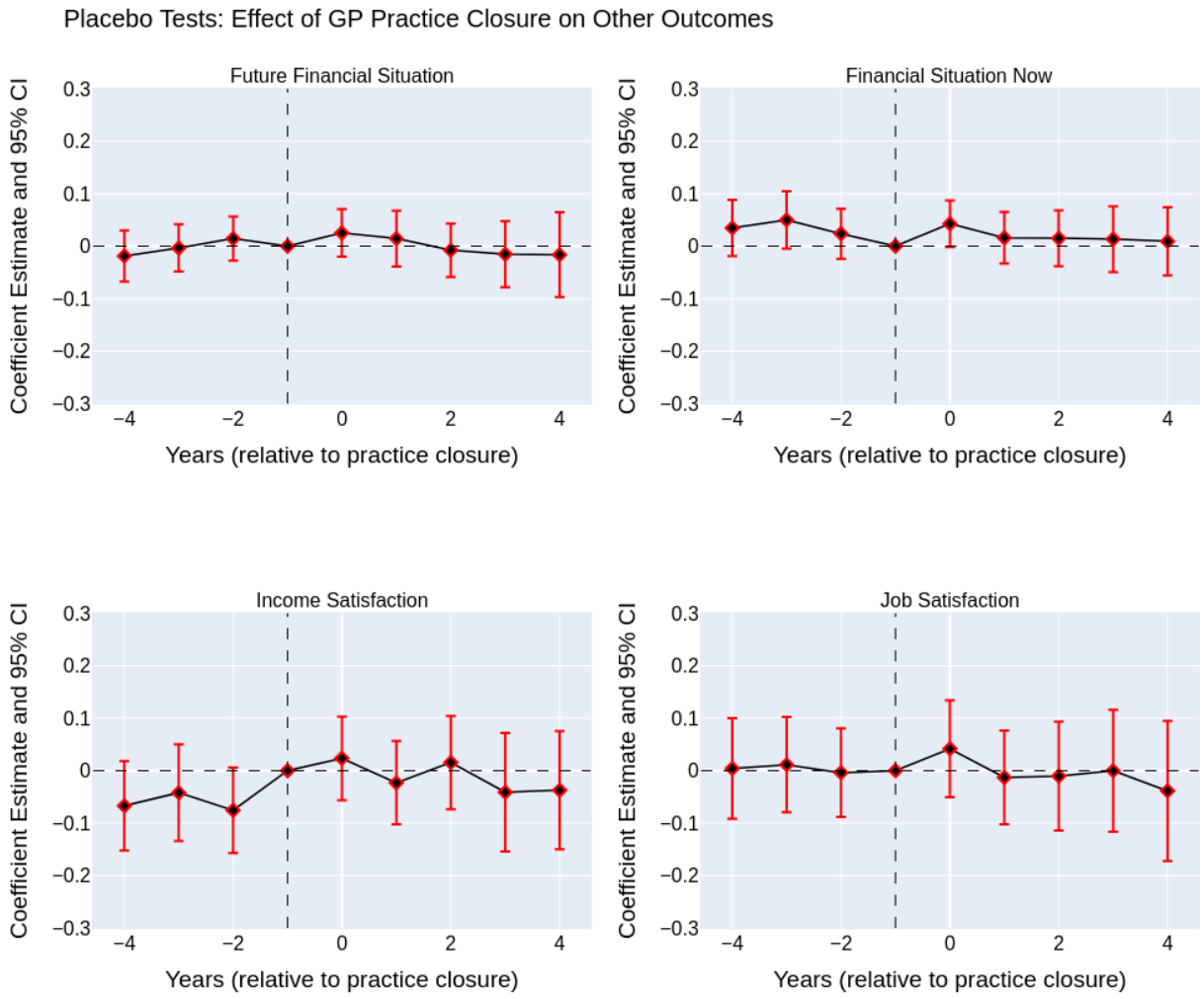


Note: Estimates and 95% confidence intervals. Standard errors clustered by ID/year. No control variables are included. Fixed effects included for individuals, years, and MSOAs.

Placebo Tests

We conduct placebo tests using a number of outcomes that are likely to be correlated with support for PRR parties but are unlikely to be directly affected by GP practice closures. Specifically, we estimate the effects of the closures on future financial situation, current financial situation, satisfaction with income and job satisfaction. We estimate effects using our primary specification in Equation A1. The results, which are presented in Figure A15, suggest that the closures do not have statistically measurable effects on any of the variables examined.

Figure A15: Effects of GP Practice Closure on Placebo Outcomes



Source: *Understanding Society (Waves 1-13)*

Note: Estimates and 95% confidence intervals. Standard errors clustered by ID/year. No control variables are included. Fixed effects included for individuals, years, and MSOAs.

Treatment Schedule

The following table presents the number of indidivuals who are 'treated' (e.g. exposed to a closure) in each year. All units that are treated before the panel begins (e.g. in the BES data, which starts in 2014) are dropped from the analysis.

Table A35: Cohort-specific Treatment Schedule (BES data)

	Cohort (first treatment year)	2015	2016	2017	2018	2019	2020	2021	2022
N	Treated	134	153	322	222	510	390	340	297
	Control	25210	24847	37755	32265	24124	46775	24499	29959

Note: Number of individuals under treatment and control for each year of the dataset.

Table A36: Cohort-specific Treatment Schedule (USOC data)

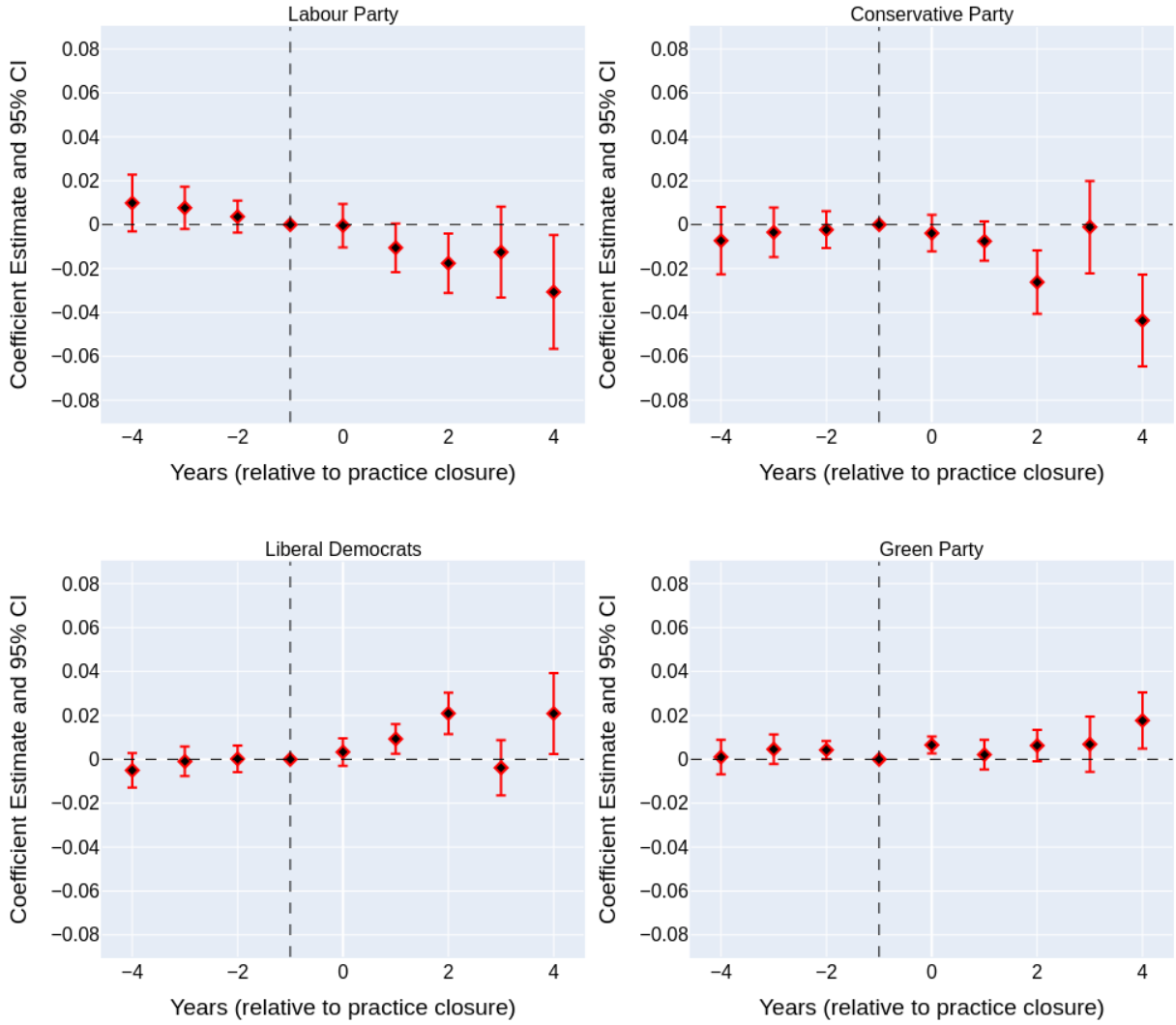
	Cohort (first treatment year)	2014	2015	2016	2017	2018	2019	2020	2021
N	Treated	15	69	66	10	83	421	283	143
	Control	15987	26560	28398	14641	12824	21151	20949	19163

Note: Number of individuals under treatment and control for each year of the dataset.

Effects of Closures on other Parties

The section provides the results of analysis on the effects of GP practice closures on other main parties. We consider the Labour Party, the Conservative Party, the Liberal Democrats and the Green Party. We present the results in two ways. Figure A16 presents event studies for each respective party. Table A37 presents the estimates in table form, with Table A38 provides the same results but with control variables included. The results suggest that voters move away from the Labour and Conservative parties, with weak evidence that they also move toward the Liberal Democrats.

Figure A16: Effects of GP Practice Closures on other Parties



Source: *British Election Study (Waves 1-25)*

Note: Estimates and 95% confidence intervals. Standard errors clustered by ID/Year. No control variables are included. Fixed effects included for individuals, years, and MSOAs.

Table A37: Event Study Results for Mainstream Party Vote Intentions

	Conservative	Labour	Lib Dem	Green
year = -5	−0.024**	0.030***	−0.005	0.004
	(0.008)	(0.007)	(0.005)	(0.004)
year = -4	−0.007	0.010	−0.005	0.001
	(0.008)	(0.007)	(0.004)	(0.004)
year = -3	−0.004	0.008	−0.001	0.005
	(0.006)	(0.005)	(0.003)	(0.003)
year = -2	−0.002	0.004	0.000	0.004*
	(0.004)	(0.004)	(0.003)	(0.002)
year = 0	−0.004	0.000	0.003	0.007***
	(0.004)	(0.005)	(0.003)	(0.002)
year = 1	−0.008	−0.011*	0.009**	0.002
	(0.005)	(0.006)	(0.003)	(0.003)
year = 2	−0.026***	−0.018**	0.021***	0.006
	(0.007)	(0.007)	(0.005)	(0.004)
year = 3	−0.001	−0.013	−0.004	0.007
	(0.011)	(0.011)	(0.006)	(0.006)
year = 4	−0.044***	−0.031**	0.021*	0.018**
	(0.011)	(0.013)	(0.009)	(0.007)
year = 5	−0.019	−0.014	0.013	−0.005
	(0.012)	(0.013)	(0.008)	(0.007)
Num.Obs.	518 092	518 092	518 092	518 092
R2	0.740	0.755	0.616	0.584
R2 Adj.	0.684	0.703	0.535	0.496
R2 Within	0.000	0.000	0.000	0.000
R2 Within Adj.	0.000	0.000	0.000	0.000
AIC	193 178.7	127 115.3	−89 368.5	−385 432.8
BIC	1 205 000.1	1 138 936.7	922 452.9	626 388.6
RMSE	0.24	0.23	0.19	0.14
FE: ID	X	X	X	X
FE: Year	X	X	X	X
FE: MSOA	X	X	X	X

Note: Standard errors clustered at the ID/Year level in parentheses. Robustness checks for the event study design, testing the effect of GP practice closures on vote intentions for all mainstream parties. Standard errors clustered by ID/year. No control variables are included in the models.

Table A38: Event Study Results for Mainstream Party Vote Intentions

	Conservative	Labour	Lib Dem	Green
year = -5	−0.027**	0.038***	−0.008	0.004
	(0.009)	(0.007)	(0.006)	(0.004)
year = -4	−0.011	0.014*	−0.008	0.004
	(0.009)	(0.007)	(0.005)	(0.004)
year = -3	−0.006	0.012**	−0.004	0.006
	(0.006)	(0.005)	(0.004)	(0.004)
year = -2	−0.004	0.007	−0.002	0.004
	(0.005)	(0.004)	(0.003)	(0.003)
year = 0	−0.004	0.000	0.000	0.007**
	(0.005)	(0.006)	(0.004)	(0.002)
year = 1	−0.003	−0.016*	0.007	0.004
	(0.005)	(0.007)	(0.004)	(0.004)
year = 2	−0.033**	−0.020**	0.025***	0.006
	(0.009)	(0.008)	(0.006)	(0.004)
year = 3	0.000	−0.029*	0.000	0.007
	(0.015)	(0.015)	(0.009)	(0.009)
year = 4	−0.071***	−0.036	0.033*	0.026**
	(0.017)	(0.021)	(0.014)	(0.010)
year = 5	−0.014	−0.010	0.012	−0.007
	(0.024)	(0.026)	(0.018)	(0.013)
IMD Score	0.007***	−0.003	−0.003*	0.001
	(0.002)	(0.002)	(0.001)	(0.001)
Immigration	1.661	0.023	−0.656	0.543
	(1.091)	(0.682)	(0.793)	(0.442)
Migrant GP Registrations	−1.887*	−0.384	0.532	−0.740
	(0.840)	(0.711)	(0.614)	(0.425)
Employment Rate	0.000	0.000*	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Num.Obs.	466 386	466 386	466 386	466 386
R2	0.745	0.765	0.628	0.592
R2 Adj.	0.688	0.712	0.544	0.500
R2 Within	0.001	0.000	0.000	0.000
R2 Within Adj.	0.001	0.000	0.000	0.000
AIC	176 698.5	98 015.0	−87 710.4	−360 570.4
BIC	1 126 739.3	1 048 055.7	862 330.3	589 470.4
RMSE	0.24	0.22	0.18	0.14
FE: ID	X	X	X	X
FE: Year	X	X	X	X
FE: MSOA	X	X	X	X

Note: Standard errors clustered by ID/Year in parentheses. All control variables included (IMD score, employment rate, international migration, and migrant GP registrations).

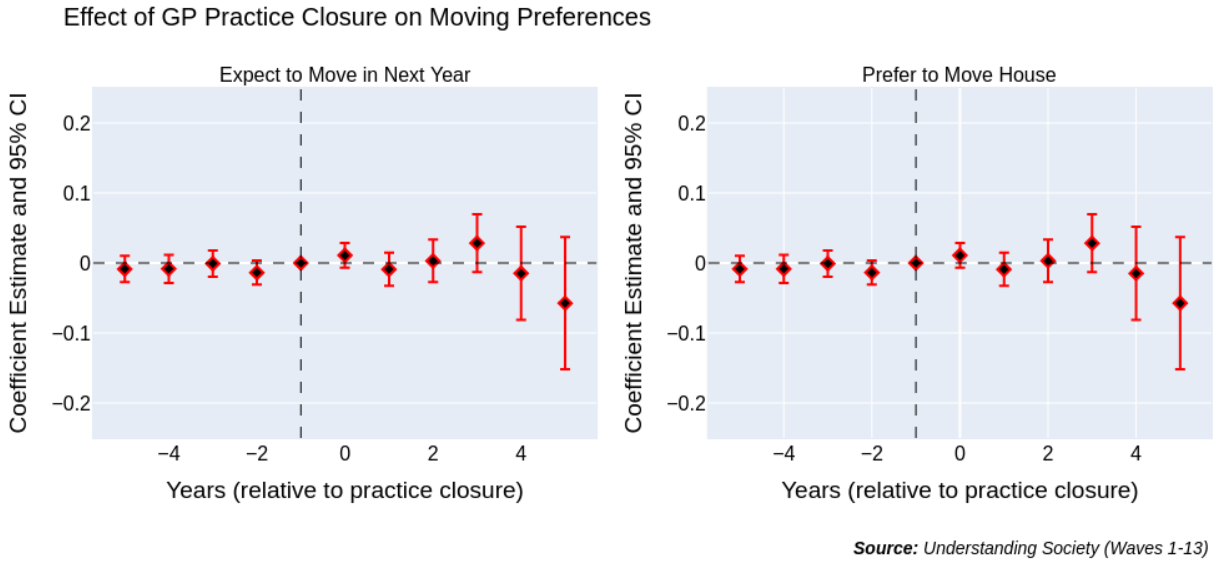
Effects of Closures on Moving

We examine the effects of the closures on preferences and expectations to move houses. There are two questions in the Understanding Society data that ask the following:

1. *If you could choose, would you stay here in your present home or would you prefer to move somewhere else?*
2. *Even though you may not want to move, do you expect you will move in the coming year*

We use these variables as outcomes to examine whether the closures lead to respondents moving to a new location. For each outcome, we create a dummy variable that is 1 when the respondent wants to, or expects to move houses. We estimate effects using our primary specification in Equation A1 and present the results as event studies in Figure A17.

Figure A17: Effects of GP Practice Closures on Moving Preferences/Expectations



Note: Estimates and 95% confidence intervals. Standard errors clustered by ID/Year. No control variables are included. Fixed effects included for individuals, years, and MSOAs.

Vote Intention vs. Actual Vote Share

In order to assess the validity of survey-based vote intention measures (from the British Election Study, BES), we compare them against the corresponding realized electoral outcomes. Let

$$I_{p,t} \quad \text{denote the average vote intention for party } p \text{ in election year } t,$$

and

$$A_{p,t} \quad \text{denote the realized vote share for the same party } p \text{ in election year } t.$$

The empirical analysis proceeds in two steps:

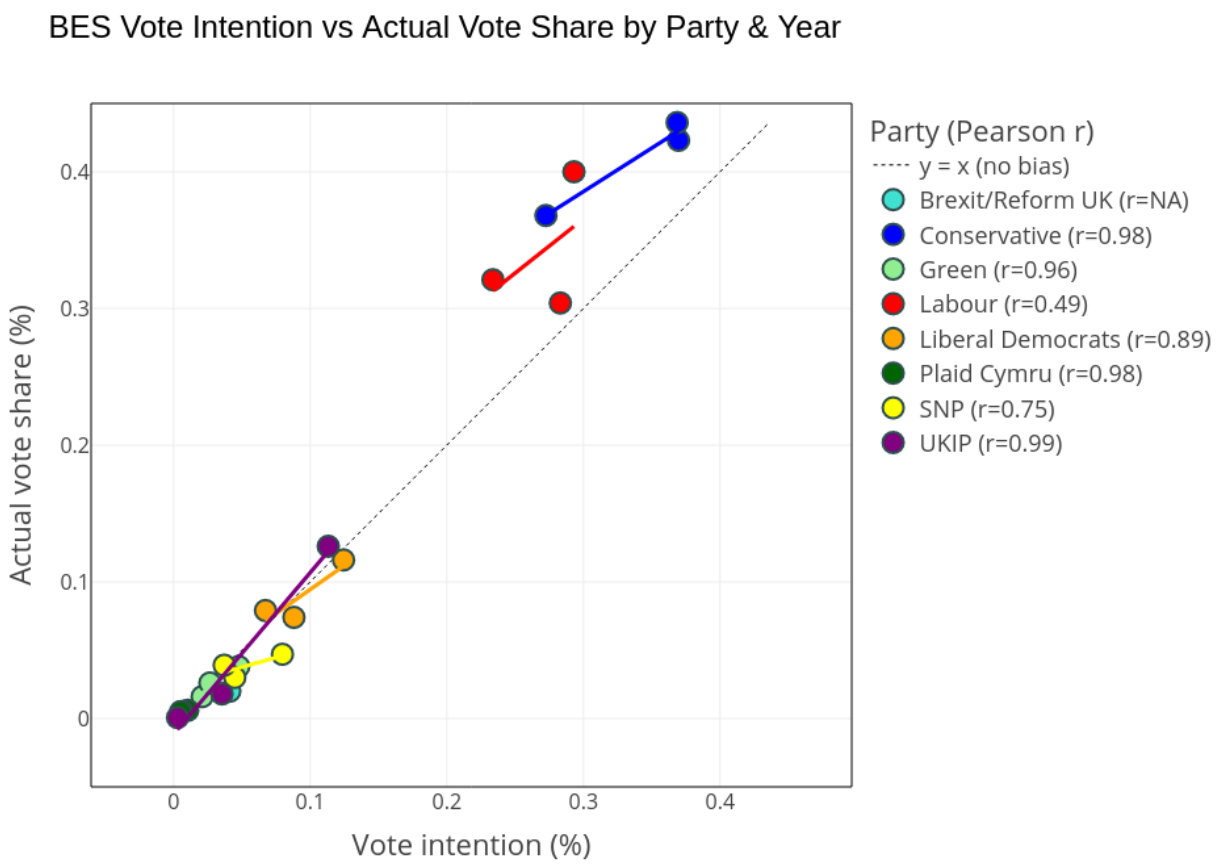
1. **Correlation analysis.** For each party p, We compute the Pearson correlation coefficient
$$r_p \;=\; \frac{\mathrm{Cov}(I_{p,\cdot}, A_{p,\cdot})}{\sigma(I_{p,\cdot})\,\sigma(A_{p,\cdot})},$$
where the covariance and standard deviations are taken over all election years t in which data are observed. This statistic quantifies the extent to which within-party variation in vote intention aligns with actual electoral outcomes across time.

2. **Scaling for visualization.** Because parties differ substantially in their absolute vote shares (e.g. major vs. minor parties), the raw scatterplot of $(I_{p,t}, A_{p,t})$ tends to be sparse and dominated by the largest parties. To facilitate visual comparison across parties, we apply a party-specific rescaling transformation. For each party p and variable $X \in \{I, A\}$, the scaled values are defined as
$$\tilde{X}_{p,t} \;=\; \frac{X_{p,t} - \min_t X_{p,t}}{\max_t X_{p,t} - \min_t X_{p,t}} \;\in [0,1],$$
i.e. a min–max normalization over time within each party. This transformation preserves the rank order and proportional differences within a party, while mapping all series onto a common unit interval.

3. **Visualization.** We then plot, for each party, the scaled vote intention $\tilde{I}_{p,t}$ against the scaled actual vote share $\tilde{A}_{p,t}$. The 45° reference line ($y = x$) indicates perfect calibration between survey intentions and electoral outcomes. Ordinary least squares (OLS) regression lines are overlaid to highlight systematic deviations from this benchmark. These figures are presented below in Figure A18 and Figure A19.

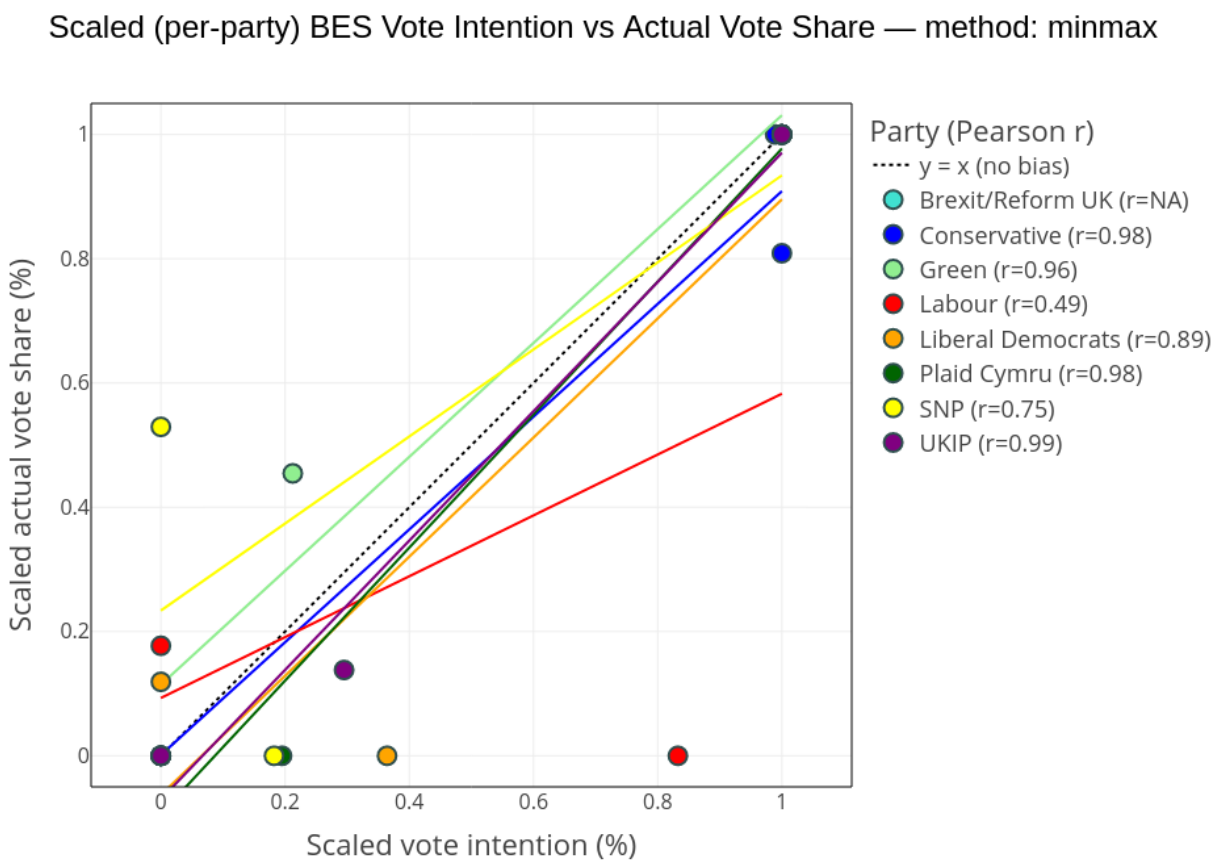
This procedure allows for both a quantitative assessment (via r_p) and a qualitative visualization of the relationship between stated vote intentions in BES surveys and realized vote shares, across multiple parties and election years.

Figure A18: Correlation between Vote Intention and Vote Behavior



Note: Correlation between vote intention measured in the BES and realized vote share for 2015, 2017 and 2019 General elections.

Figure A19: Correlation between Vote Intention and Vote Behavior (Scaled)



Note: Correlation (min-max scaled) between vote intention measured in the BES and realized vote share for 2015, 2017 and 2019 General elections.

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